

Chapter 5

Connectivity

5.1 Cut-Vertices

It is probably clear by now that one of the most important properties that a graph can possess is that of being connected. Figure 5.1 shows seven graphs of order 7. The graph G_1 is a tree, $G_4 \cong C_7$, and $G_7 \cong K_7$. Obviously, all of these graphs are connected. However, some appear to be “more connected” than others. This statement suggests the possibility of introducing a measure of how connected a graph is, which is the goal of this chapter.

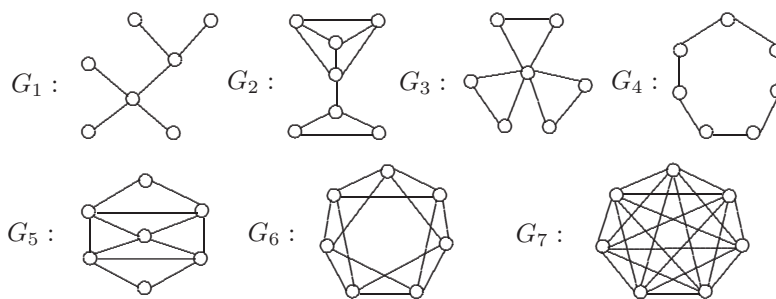


Figure 5.1: Connected graphs

Some graphs are so slightly connected that the removal of a single edge results in a disconnected graph. We have already seen this and an edge with this property is a bridge. The graph G_2 has a bridge. So does G_1 . In fact, *every* edge of G_1 is a bridge since G_1 is a tree. We now turn from connected graphs containing an edge whose removal results in a disconnected graph to connected graphs containing a *vertex* whose removal results in a disconnected graph.

Recall that if v is a vertex of a nontrivial graph G , then by $G - v$ we mean the (induced) subgraph of G whose vertex set consists of all vertices of G except v

and whose edge set consists of all edges of G except those incident with v . This concept is illustrated in Figure 5.2. In fact, if U is a proper subset of the vertex set of G , then $G - U$ is the (induced) subgraph of G whose vertex set is $V(G) - U$ and whose edge set consists of all edges of G joining two vertices in $V(G) - U$. A vertex v in a connected graph G is a **cut-vertex** of G if $G - v$ is disconnected. More generally, a vertex v is a cut-vertex in a graph G if v is a cut-vertex of a component of G . In the graph G of Figure 5.2, v and s are the only cut-vertices. In the graph $G - v$, the vertex x is not a cut-vertex; however, s is a cut-vertex of $G - v$. Consequently, for $U = \{s, v\}$, the graph $G - U$ is disconnected. The graphs G_1, G_2 , and G_3 of Figure 5.1 also contain cut-vertices, but no other graphs in Figure 5.1 contain cut-vertices.

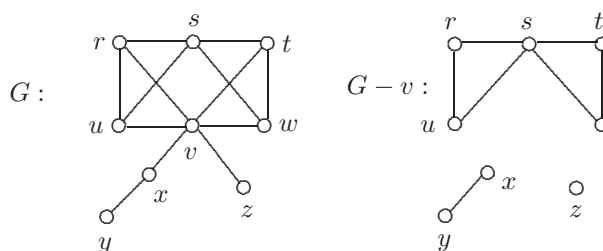


Figure 5.2: The graphs G and $G - v$

Notice that the graph G of Figure 5.2 not only contains the cut-vertex v , it contains three bridges, two of which are incident with v . We have already noticed that the two graphs of Figure 5.1 with bridges, namely G_1 and G_2 , contain cut-vertices as well (although G_3 contains a cut-vertex but no bridges). In fact, except for the graph K_2 , every connected graph with bridges contains cut-vertices as well. Recall by Theorem 4.1 that an edge e in a graph G is a bridge if and only if e lies on no cycle of G .

We now establish some facts about cut-vertices. Since a cut-vertex in a graph G is a cut-vertex of a component of G , we restrict ourselves to connected graphs.

Theorem 5.1 *Let v be a vertex incident with a bridge in a connected graph G . Then v is a cut-vertex of G if and only if $\deg v \geq 2$.*

Proof. [proof by contrapositive, proof by contradiction] Suppose that uv is a bridge of G . Then $\deg v \geq 1$. Assume that $\deg v = 1$. Since v is an end-vertex of G , the graph $G - v$ is connected and so v is not a cut-vertex of G .

For the converse, assume that $\deg v \geq 2$. Then there is a vertex w different from u that is adjacent to v . Assume, to the contrary, that v is not a cut-vertex. Thus $G - v$ is connected, and so there is a $u - w$ path P in $G - v$. However then, P together with v and the two edges uv and vw form a cycle containing the bridge uv . This contradicts Theorem 4.1. ■

One immediate consequence of Theorem 5.1 is that if a vertex v of a nontrivial tree T is not an end-vertex of T , then v is a cut-vertex of T . Another immediate consequence of Theorem 5.1 is stated next.

Corollary 5.2 *Let G be a connected graph of order 3 or more. If G contains a bridge, then G contains a cut-vertex.*

If v is a cut-vertex in a connected graph G , then, of course, $G - v$ contains two or more components. If u and w are vertices in distinct components of $G - v$, then u and w are not connected in $G - v$. On the other hand, u and w are necessarily connected in G . These observations lead us to the following theorem.

Theorem 5.3 *Let v be a cut-vertex in a connected graph G , and let u and w be vertices in distinct components of $G - v$. Then v lies on every $u - w$ path in G .*

We can now present a characterization of vertices in a connected graph G that are cut-vertices of G .

Corollary 5.4 *A vertex v of a connected graph G is a cut-vertex of G if and only if there exist vertices u and w distinct from v such that v lies on every $u - w$ path of G .*

Proof. [direct proof] Suppose that v is a cut-vertex of G . Then $G - v$ is disconnected. Let u and w be vertices in different components of $G - v$. It then follows by Theorem 5.3 that every $u - w$ path in G contains v .

On the other hand, if G contains two vertices u and w such that every $u - w$ path in G contains v , then there is no $u - w$ path in $G - v$. Thus u and w are not connected in $G - v$, and so $G - v$ is disconnected. Therefore, v is a cut-vertex of G . ■

By Corollary 5.4, if v is a vertex in a connected graph G that is not a cut-vertex of G , then for *every* two vertices u and w of G that are distinct from v , there is a $u - w$ path that does not contain v .

While the concepts of bridges and cut-vertices are parallel concepts and have a number of similarities, they have some major differences as well. We have seen that it's possible for every edge of a connected graph G to be a bridge. Of course, G must then be a tree. On the other hand, every nontrivial connected graph must contain vertices that are not cut-vertices.

Theorem 5.5 *Let G be a nontrivial connected graph and let $u \in V(G)$. If v is a vertex that is farthest from u in G , then v is not a cut-vertex of G .*

Proof. [proof by contradiction] Assume, to the contrary, that v is a cut-vertex of G . Let w be a vertex belonging to a component of $G - v$ that does not contain u . Since every $u - w$ path contains v , it follows that $d(u, w) > d(u, v)$, which is a contradiction. ■

It is now immediate that every nontrivial connected graph contains at least *two* vertices that are not cut-vertices.

Corollary 5.6 *Every nontrivial connected graph contains at least two vertices that are not cut-vertices.*

Proof. [direct proof] Let u and v be vertices of a nontrivial connected graph G such that $d(u, v) = \text{diam}(G)$. Since each of u and v is farthest from the other, it follows by Theorem 5.5 that both u and v are not cut-vertices of G . ■

Exercises for Section 5.1

5.1 Give an example of a graph that

- (a) contains more bridges than cut-vertices.
- (b) contains more cut-vertices than bridges.

5.2 (a) For each integer $k \geq 2$, give an example of a connected graph G containing a vertex v such that $G - v$ has k components.

- (b) Give an example of a connected graph G of order 3 or more containing vertices u and v such that $G - u - v$ has fewer components than $G - u$.

5.3 Prove or disprove:

- (a) If a vertex v of a graph G lies on a cycle of G , then v is not a cut-vertex.
- (b) If a vertex v of a graph G does not lie on any cycle of G , then v is a cut-vertex.
- (c) A tree of order 3 or more has more cut-vertices than end-vertices.
- (d) A tree of order 3 or more has more cut-vertices than bridges.

5.4 Prove that if v is a cut-vertex of a graph G , then v is not a cut-vertex of the complement $\overline{G}$ of G .

5.5 Find a counterexample to each of the following statements.

- (a) If G is a connected graph of order 13 and v is a cut-vertex of G , then there exists a component of $G - v$ containing at least 7 vertices.
- (b) If G is a connected graph containing only even vertices, then G contains no cut-vertices.
- (c) If G is a connected graph with a cut-vertex, then G contains a bridge.
- (d) If G is a connected graph with a bridge, then G contains a cut-vertex.

5.6 Prove that a 3-regular graph G has a cut-vertex if and only if G has a bridge.

- 5.7 Prove that if T is a tree of order at least 3, then T contains a cut-vertex v such that every vertex adjacent to v , with at most one exception, is an end-vertex.
- 5.8 (a) Let G be a nontrivial connected graph. Prove that if v is an end-vertex of a spanning tree of G , then v is not a cut-vertex of G .
- (b) Use (a) to give an alternative proof of the fact that every nontrivial connected graph contains at least two vertices that are not cut-vertices.
- (c) Let v be a vertex in a nontrivial connected graph G . Show that there exists a spanning tree of G that contains all edges of G that are incident with v .
- (d) Prove that if a connected graph G has exactly two vertices that are not cut-vertices, then G is a path. [Recall that if a tree contains a vertex of degree exceeding 2, then T has more than two end-vertices.]

5.2 Blocks

We now turn our attention from connected graphs containing cut-vertices to connected graphs that contain no cut-vertices. A nontrivial connected graph with no cut-vertices is called a **nonseparable graph**. Hence all of the graphs G_4 , G_5 , G_6 , and G_7 of Figure 5.1 are nonseparable. In addition, K_2 is a nonseparable graph; indeed, K_2 is the only nonseparable graph of order 2. Since nonseparable graphs of order 3 or more contain no cut-vertices, they contain no bridges; that is, every edge lies on a cycle. In fact, more can be said.

Theorem 5.7 *A graph of order at least 3 is nonseparable if and only if every two vertices lie on a common cycle.*

Proof. [proof by contradiction] First, suppose that G is a graph of order at least 3 such that every two vertices of G lie on a common cycle. Assume, to the contrary, that G is not nonseparable. Since every two vertices lie on a common cycle, every two vertices are connected and so G is connected. Consequently, G must contain a cut-vertex, say v . Let u and w be two vertices that belong to different components of $G - v$. By assumption, u and w lie on a common cycle C on G . However then, C determines two distinct $u - w$ paths of G , at least one of which does not contain v , contradicting Theorem 5.3. Therefore, G contains no cut-vertices and so G is nonseparable.

We now verify the converse. Let G be a nonseparable graph of order at least 3 and assume, to the contrary, that there are pairs of vertices of G that do not lie on a common cycle. Among all such pairs, let u, v be a pair for which $d(u, v)$

is minimum. If $d(u, v) = 1$, then $uv \in E(G)$ and uv must lie on a cycle, as we observed earlier. Therefore, $d(u, v) = k \geq 2$.

Let $P : u = v_0, v_1, \dots, v_{k-1}, v_k = v$ be a $u - v$ path of length k in G . Since $d(u, v_{k-1}) = k - 1 < k$, there is a cycle C containing u and v_{k-1} . By assumption, v is not on C . Since v_{k-1} is not a cut-vertex of G and u and v are distinct from v_{k-1} , it follows by Theorem 5.4 that there is a $v - u$ path Q that does not contain v_{k-1} . Since u is on C , there is a first vertex of Q that is on C . Let x be that vertex. Let Q' be the $v - x$ subpath of Q (see Figure 5.3) and let P' be a $v_{k-1} - x$ path on C that contains u . (If $x \neq u$, then the path P' is unique.) However, the cycle C' produced by proceeding from v to its neighbor v_{k-1} , along P' to x , and then along Q' to v contains both u and v , a contradiction. ■

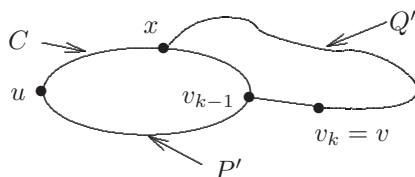


Figure 5.3: The cycle C and paths P' , Q' in the proof of Theorem 5.7

If a nontrivial connected graph is not nonseparable, then it must contain cut-vertices as well as certain nonseparable subgraphs that are of special interest. A nonseparable subgraph of a graph G that is not a proper subgraph of any other nonseparable subgraph in G is called a **block** of G . Each block of G is an induced subgraph of G . Therefore, if G itself is nonseparable, then G has only one block, namely, the graph G . On the other hand, if G is connected and has cut-vertices, then G has two or more blocks. Figure 5.4 shows a connected graph G with two cut-vertices u and v and four blocks B_1 , B_2 , B_3 , and B_4 . So, just as a disconnected graph has special connected subgraphs called components, a connected graph with cut-vertices contains special nonseparable subgraphs called blocks.

There is another way to look at blocks. First, we describe an equivalence relation that is defined on the edge set of a nontrivial connected graph. (Again, recall that equivalence relations are reviewed in Appendix 2.)

Theorem 5.8 *Let R be the relation defined on the edge set of a nontrivial connected graph G by $e R f$, where $e, f \in E(G)$, if $e = f$ or e and f lie on a common cycle of G . Then R is an equivalence relation.*

Proof. [direct proof] It is immediate that R is reflexive and symmetric, so we need only show that R is transitive. Let $e, f, g \in E(G)$ such that $e R f$ and $f R g$. If $e = f$ or $f = g$, then it follows that $e R g$. Hence we may assume that e and f lie on a cycle C and that f and g lie on a cycle C' . If e lies on C' or g lies on C , then $e R g$. Thus we may assume that this does not occur.

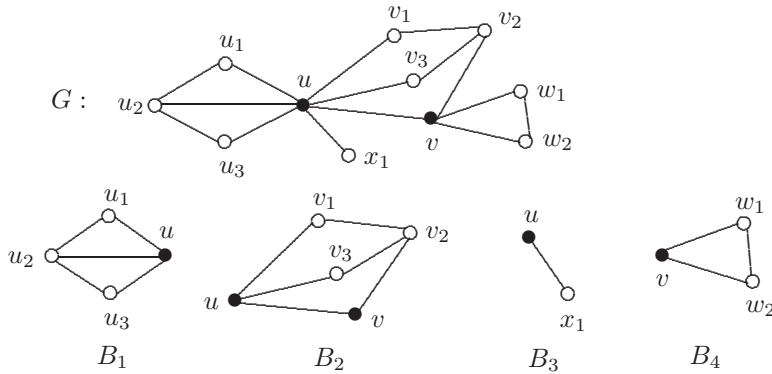


Figure 5.4: A connected graph and its blocks

Let $e = uv$ and suppose that P is the $u - v$ path on C not containing e . Let x be the first vertex of P belonging to C' and y the last vertex of P belonging to C' . Furthermore, let P' be the $x - y$ path on C' that contains g and let P'' be the $x - y$ path on C that contains e . Then P' and P'' produce a cycle C'' containing both e and g . Therefore, $e R g$. ■

The equivalence relation R described in Theorem 5.8 produces a partition of the edge set of every nontrivial connected graph G into equivalence classes. Each subgraph of G induced by the edges in an equivalence class is in fact a block of G . The following corollary of Theorem 5.8 provides properties of blocks in a nontrivial connected graph.

Corollary 5.9 *Every two distinct blocks B_1 and B_2 in a nontrivial connected graph G have the following properties:*

- (a) *The blocks B_1 and B_2 are edge-disjoint.*
- (b) *The blocks B_1 and B_2 have at most one vertex in common.*
- (c) *If B_1 and B_2 have a vertex v in common, then v is a cut-vertex of G .*

Proof. [direct proof, proof by contradiction] That every two distinct blocks are edge-disjoint is an immediate consequence of Theorem 5.8. We now verify (b). Assume, to the contrary, that B_1 and B_2 have two distinct vertices u and v in common. Since B_1 and B_2 are connected subgraphs of G , there is a $u - v$ path P' in B_1 and a $u - v$ path P'' in B_2 . Furthermore, since B_1 and B_2 are edge-disjoint, so too are P' and P'' . Let w be the first vertex that P' and P'' have in common after u (possibly $w = v$). The $u - w$ subpath Q' of P' and $u - w$ subpath Q'' of P'' form a cycle in G containing an edge e_1 of B_1 and an edge e_2 of B_2 . Hence e_1 and e_2 belong to the same block of G , which is impossible. This verifies (b).

It remains to verify (c). Suppose that two blocks B_1 and B_2 of G have a vertex v in common. Then v is incident with an edge $e_1 = vv_1$ in B_1 and an edge $e_2 = vv_2$ in B_2 . Assume, to the contrary, that v is not a cut-vertex of G . By Corollary 5.4, G has a $v_1 - v_2$ path P not containing v . Then P together with v and the edges e_1 and e_2 produce a cycle containing e_1 and e_2 . This is impossible, however, since e_1 and e_2 belong to distinct blocks of G . ■

We have indicated two ways that blocks can be described. Exercise 5.15 asks you to show that these two interpretations of blocks are equivalent.

Exercises for Section 5.2

- 5.9 For the graph G of Figure 5.5, determine the cut-vertices, bridges, and blocks of G .

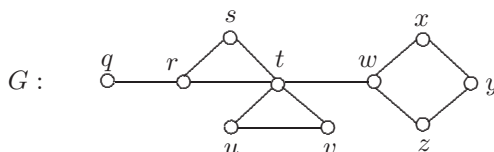


Figure 5.5: The graph G in Exercise 5.9

- 5.10 Prove that a connected graph G of size at least 2 is nonseparable if and only if any two adjacent edges of G lie on a common cycle of G .
- 5.11 Prove that if G is a graph of order $n \geq 3$ such that $\deg v \geq n/2$ for every vertex v of G , then G is nonseparable.
- 5.12 If a connected graph G contains three blocks and k cut-vertices, what are the possible values for k ? Explain your answer.
- 5.13 Prove or disprove: If G is a connected graph with cut-vertices and u and v are vertices of G such that $d(u, v) = \text{diam}(G)$, then no block of G contains both u and v .
- 5.14 Let G be a connected graph containing a cut-vertex v and let G_1 be a component of $G - v$.
- Show that the induced subgraph $\langle V(G_1) \cup \{v\} \rangle$ of G is connected.
 - Show that the induced subgraph $\langle V(G_1) \cup \{v\} \rangle$ of G need not be a block of G .
- 5.15 For a nontrivial connected graph G , a block of G has been defined as (1) a nonseparable subgraph of G that is not a proper subgraph of any other nonseparable subgraph of G and has been described as (2) a subgraph of G

induced by the edges in an equivalence class resulting from the equivalence relation defined in Theorem 5.8. Show that these two interpretations of blocks are equivalent.

5.16 Give an example of a graph G with the following properties:

- (a) every two adjacent vertices lie on a common cycle,
- (b) there exist two adjacent edges that do not lie on a common cycle.

5.3 Connectivity

A connected graph G with cut-vertices has the property that G can be disconnected by the removal of a single vertex. On the other hand, nonseparable graphs contain no cut-vertices. In a sense, then, nonseparable graphs are more highly connected than connected graphs with cut-vertices. This suggests a way of measuring the connectedness of graphs. By a **vertex-cut** in a graph G , we mean a set U of vertices of G such that $G - U$ is disconnected. A vertex-cut of minimum cardinality in G is called a **minimum vertex-cut**. If G is not complete, then G contains two nonadjacent vertices. The removal of all vertices of G except these two nonadjacent vertices produces a disconnected graph. In other words, every graph that is not complete has a vertex-cut. On the other hand, the removal of any proper subset of vertices of a complete graph results in another complete graph. Therefore, a connected graph contains a vertex-cut if and only if G is not complete.

If U is a minimum vertex-cut in a noncomplete connected graph G , then $G - U$ is disconnected and contains components $G_1, G_2, \dots, G_k$, where $k \geq 2$. Every vertex $u \in U$ is adjacent to at least one vertex in G_i for each i ($1 \leq i \leq k$); for otherwise $U - \{u\}$ is also a vertex-cut, which is impossible. The structure of such a graph G is indicated in Figure 5.6, where there are no edges between any two distinct components of $G - U$.

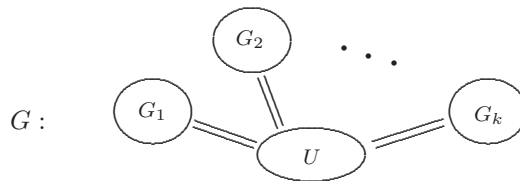


Figure 5.6: A minimum vertex-cut in a graph

For a graph G that is not complete, the **vertex-connectivity** (or simply the **connectivity**) $\kappa(G)$ of G is defined as the cardinality of a minimum vertex-cut

of G ; if $G \cong K_n$ for some positive integer n , then $\kappa(G)$ is defined to be $n - 1$. (The symbol κ is the Greek letter kappa.) In general then, the **connectivity** $\kappa(G)$ of a graph G is the minimum value of $|U|$ among all subsets U of $V(G)$ such that $G - U$ is either disconnected or trivial. Therefore, for every graph G of order n ,

$$0 \leq \kappa(G) \leq n - 1.$$

Thus a nontrivial graph G has connectivity 0 if and only if G is disconnected; a graph G has connectivity 1 if and only if $G \cong K_2$ or G is a connected graph with cut-vertices; and a graph G has connectivity 2 or more if and only if G is a nonseparable graph of order 3 or more. For the graphs of Figure 5.1, $\kappa(G_1) = \kappa(G_2) = \kappa(G_3) = 1$, $\kappa(G_4) = \kappa(G_5) = 2$, $\kappa(G_6) = 4$, and $\kappa(G_7) = 6$.

As it turns out, we will often be more interested in graphs that cannot be disconnected by removing some prescribed number of vertices. For a nonnegative integer k , a graph G is said to be **k -connected** if $\kappa(G) \geq k$. Therefore, a k -connected graph is also ℓ -connected for every integer ℓ with $0 \leq \ell \leq k$. In particular, the graphs G_4, G_5, G_6 , and G_7 of Figure 5.1 are all 2-connected. The graphs G_6 and G_7 are also 3-connected, although neither G_4 nor G_5 is 3-connected. Thus G is 1-connected if and only if G is nontrivial and connected; while G is 2-connected if and only if G is nonseparable and has order at least 3. In general, a graph G is k -connected if and only if the removal of fewer than k vertices does not result in a disconnected or trivial graph.

The connectivity of a graph G provides a measure of “how connected” G is. There are other measures, including a common one involving the edges of G . An **edge-cut** in a nontrivial graph G is a set X of edges of G such that $G - X$ is disconnected. An edge-cut X of a connected graph G is **minimal** if no proper subset of X is an edge-cut of G . If X is a minimal edge-cut of a connected graph G , then $G - X$ contains exactly two components G_1 and G_2 . Necessarily then, X consists of all those edges of G joining G_1 and G_2 .

If X is an edge-cut of a connected graph G that is not minimal, then there is a proper subset Y of X that is a minimal edge-cut. An edge-cut of minimum cardinality is called a **minimum edge-cut**. While every minimum edge-cut is a minimal edge-cut, the converse is not true. For the graph H of Figure 5.7, consider the sets $X_1 = \{e_3, e_4, e_5\}$, $X_2 = \{e_1, e_2, e_6\}$, and $X_3 = \{e_1, e_6\}$ of edges of H . All three of these sets are edge-cuts because all of the graphs $H - X_1$, $H - X_2$, and $H - X_3$ are disconnected. Both X_1 and X_3 are minimal edge-cuts, while X_2 is not a minimal edge-cut as X_3 is a proper subset of X_2 . The set X_3 is a minimum edge-cut, while X_1 and X_2 are not.

The **edge-connectivity** $\lambda(G)$ of a nontrivial graph G is the cardinality of a minimum edge-cut of G , while we define $\lambda(K_1) = 0$. Thus $\lambda(G)$ is the minimum value of $|X|$ among all subsets X of $E(G)$ such that $G - X$ is either a disconnected or trivial graph. For every graph G of order n ,

$$0 \leq \lambda(G) \leq n - 1.$$

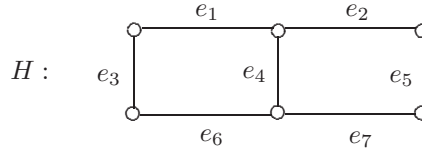


Figure 5.7: Illustrating edge-cuts in a graph

Note that $\lambda(G) = 0$ if and only if G is disconnected or G is trivial, while $\lambda(G) = 1$ if and only if G is connected and contains a bridge. For the graph H of Figure 5.7, $\lambda(G) = 2$ since $X_3 = \{e_1, e_6\}$ is a minimum edge-cut in H .

For a nonnegative integer k , a graph G is **k -edge-connected** if $\lambda(G) \geq k$. Consequently, every k -edge-connected graph is ℓ -edge-connected for every integer ℓ with $0 \leq \ell \leq k$. Therefore, every 1-edge-connected graph is nontrivial and connected and every 2-edge-connected graph is a connected graph of order 3 or more that contains no bridges. For example, the graphs G_3 , G_4 , and G_5 of Figure 5.1 are 2-edge-connected graphs that are not 3-edge-connected.

Example 5.10 Show that $\lambda(K_n) = n - 1$.

Solution. By definition, $\lambda(K_1) = 0$. Let $G \cong K_n$ for $n \geq 2$. Since every vertex of G has degree $n - 1$, if we remove the $n - 1$ edges incident with a vertex, then a disconnected graph results. Thus $\lambda(G) \leq n - 1$. Now let X be a minimum edge-cut of G . So $|X| = \lambda(G)$. Then $G - X$ has exactly two components G_1 and G_2 , where G_1 has order k , say, and G_2 has order $n - k$. Since X consists of all edges joining G_1 and G_2 , and G is complete, it follows that $|X| = k(n - k)$. Because $k \geq 1$ and $n - k \geq 1$, we have $(k - 1)(n - k - 1) \geq 0$ and so

$$(k - 1)(n - k - 1) = k(n - k) - n + 1 \geq 0.$$

Hence $\lambda(G) = |X| = k(n - k) \geq n - 1$. Therefore, $\lambda(K_n) = n - 1$. $\diamond$

As we noted earlier, complete graphs do not contain vertex-cuts. But for a graph G that is not complete, the cardinality of a minimum vertex-cut in G can never exceed the cardinality of any edge-cut in G . Indeed, the following theorem provides us with inequalities concerning the connectivity, edge-connectivity, and minimum degree of a graph. The proof is similar, in part, to the argument we just used to show that $\lambda(K_n) = n - 1$ in Example 5.10.

Theorem 5.11 For every graph G ,

$$\kappa(G) \leq \lambda(G) \leq \delta(G).$$

Proof. [direct proof] If G is disconnected or trivial, then $\kappa(G) = \lambda(G) = 0$ and the inequalities hold; while if $G \cong K_n$ for some integer $n \geq 2$, then $\kappa(G) = \lambda(G) = \delta(G) = n - 1$. Thus we may assume that G is a connected graph of order $n \geq 3$ that is not complete. Hence $\delta(G) \leq n - 2$.

First, we show that $\lambda(G) \leq \delta(G)$. Let v be a vertex of G with $\deg v = \delta(G)$. Since the set of the $\delta(G)$ edges incident with v in G is an edge-cut of G , it follows that

$$\lambda(G) \leq \delta(G) \leq n - 2.$$

It remains to show that $\kappa(G) \leq \lambda(G)$. Let X be a minimum edge-cut of G . Then $|X| = \lambda(G) \leq n - 2$. Necessarily, $G - X$ contains exactly two components G_1 and G_2 . Suppose that the order of G_1 is k . Thus the order of G_2 is $n - k$, where $k \geq 1$ and $n - k \geq 1$. Consequently, every edge in X joins a vertex of G_1 and a vertex of G_2 . We consider two cases.

Case 1. Every vertex of G_1 is adjacent in G to every vertex of G_2 . Thus $|X| = k(n - k)$. Since $(k - 1)(n - k - 1) \geq 0$, it follows that

$$(k - 1)(n - k - 1) = k(n - k) - n + 1 \geq 0$$

and so $\lambda(G) = |X| = k(n - k) \geq n - 1$. However, $\lambda(G) \leq n - 2$; so this case cannot occur.

Case 2. There exist vertices u in G_1 and v in G_2 such that u and v are not adjacent in G . We now define a set U of vertices of G . For each $e \in X$, we select a vertex for U in the following way. If u is incident with e , then choose the other vertex in G_2 that is incident with e as an element of U ; otherwise, select the vertex that is incident with e and belongs to G_1 as an element of U . Then $|U| \leq |X|$. Since $u, v \notin U$ and there is no $u - v$ path in $G - U$, it follows that $G - U$ is disconnected, and so U is a vertex-cut. Hence

$$\kappa(G) \leq |U| \leq |X| = \lambda(G),$$

as desired. ■

Theorem 5.11 is due to Hassler Whitney. Although a mathematician who made a number of important contributions to graph theory, Whitney was primarily interested in topology. Whitney was born in New York on March 23, 1907 and received his Ph.D. from Harvard in 1932 under the direction of George David Birkhoff. Whitney's doctoral dissertation was written in graph theory. He stayed on at Harvard until 1952, when he accepted an offer from the Institute for Advanced Study at Princeton. He remained there until he retired in 1977. During his life he was the recipient of many mathematical awards: the National Medal of Science (1976), the Wolf Prize (1983), and the Steel Prize (1985).

Although research was a large part of Whitney's professional life, he contributed to mathematics in many ways. During 1944–1949 he edited the *American Journal of Mathematics* and during 1949–1954 he edited *Mathematical Reviews*. During 1953–1956, he chaired the National Science Foundation mathematical panel. On the personal side, Whitney was an avid mountain climber. In fact, the Whitney-Gilman Ridge on Cannon Cliff in Franconia, New Hampshire was named for him and his cousin, who were the first to climb it (on August 3, 1939). Whitney died on May 10, 1989.

Both inequalities in Theorem 5.11 can be strict as the graph G in Figure 5.8 shows, where $\kappa(G) = 1$, $\lambda(G) = 2$, and $\delta(G) = 3$. For cubic graphs, however, the connectivity and edge-connectivity are always equal.

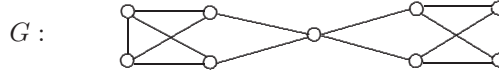


Figure 5.8: A graph G with $\kappa(G) = 1$, $\lambda(G) = 2$, and $\delta(G) = 3$

Theorem 5.12 *If G is a cubic graph, then $\kappa(G) = \lambda(G)$.*

Proof. [direct proof] For a cubic graph G , it follows that $\kappa(G) = \lambda(G) = 0$ if and only if G is disconnected. If $\kappa(G) = 3$, then $\lambda(G) = 3$ by Theorem 5.11. So two cases remain, namely $\kappa(G) = 1$ or $\kappa(G) = 2$. Let U be a minimum vertex-cut of G . Then $|U| = 1$ or $|U| = 2$. So $G - U$ is disconnected. Let G_1 and G_2 be two components of $G - U$. Since G is cubic, for each $u \in U$, at least one of G_1 and G_2 contains exactly one neighbor of u .

Case 1. $\kappa(G) = |U| = 1$. Thus U consists of a cut-vertex u of G . Since some component of $G - U$ contains exactly one neighbor w of u , the edge uw is a bridge of G and so $\lambda(G) = \kappa(G) = 1$.

Case 2. $\kappa(G) = |U| = 2$. Let $U = \{u, v\}$. Assume that each of u and v has exactly one neighbor, say u' and v' , respectively, in the same component of $G - U$. (This is the case that holds if $uv \in E(G)$.) Then $X = \{uu', vv'\}$ is an edge-cut of G and $\lambda(G) = \kappa(G) = 2$. (See Figure 5.9(a) for the situation when u and v are not adjacent.)

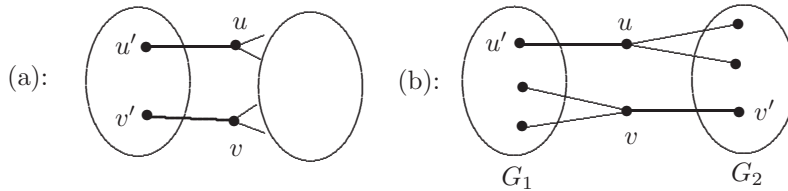


Figure 5.9: A step in the proof of Case 2

Hence we may assume that u has one neighbor u' in G_1 and two neighbors in G_2 ; while v has two neighbors in G_1 and one neighbor v' in G_2 (see Figure 5.9(b)). Therefore, $uv \notin E(G)$ and $X = \{uu', vv'\}$ is an edge-cut of G ; so $\lambda(G) = \kappa(G) = 2$. ■

The connectivity of a graph G of a given order n and size m can only be so large. For example, if $m < n - 1$, then G is disconnected by Theorem 4.7 and so $\kappa(G) = 0$. If $m \geq n - 1$, then there is a sharp upper bound for $\kappa(G)$, which we present next. (See Appendix 1 for a review of the floor and ceiling of a real number.)

Theorem 5.13 *If G is a graph of order n and size $m \geq n - 1$, then*

$$\kappa(G) \leq \left\lfloor \frac{2m}{n} \right\rfloor.$$

Proof. [direct proof] Since the sum of the degrees of the vertices of G is $2m$, the average degree of the vertices of G is $2m/n$ and so $\delta(G) \leq 2m/n$. Since $\delta(G)$ is an integer, $\delta(G) \leq \lfloor 2m/n \rfloor$. By Theorem 5.11, $\kappa(G) \leq \lfloor 2m/n \rfloor$. ■

The bound given in Theorem 5.13 is sharp in the sense that for every two integers n and m with $1 \leq n - 1 \leq m \leq \binom{n}{2}$, there exists a graph G of order n and size m such that $\kappa(G) = \lfloor \frac{2m}{n} \rfloor$. We now see why this is so. If $m = n - 1$, then every tree T of order n has the desired property as

$$\kappa(T) = \left\lfloor \frac{2m}{n} \right\rfloor = \left\lfloor \frac{2n - 2}{n} \right\rfloor = 1.$$

Hence we may assume that $3 \leq n \leq m \leq \binom{n}{2}$.

It is useful to describe a class of graphs $H_{r,n}$ for integers r and n with $2 \leq r < n$ such that $H_{r,n}$ has order n and $\kappa(H_{r,n}) = r$. Let $V(H_{r,n}) = \{v_1, v_2, \dots, v_n\}$. Suppose first that $r = 2k$ is even. For each integer i ($1 \leq i \leq n$), the vertex v_i is adjacent to $v_{i+1}, v_{i+2}, \dots, v_{i+k}$ and to $v_{i-1}, v_{i-2}, \dots, v_{i-k}$. Thus $H_{r,n}$ is an r -regular graph of order n .

Next, suppose that $r = 2k + 1$ is odd and $n = 2\ell$ is even. For each integer i ($1 \leq i \leq n$), the vertex v_i is joined to the $2k$ vertices mentioned above as well as to $v_{i+\ell}$. Here too $H_{r,n}$ is an r -regular graph of order n . (The graphs $H_{r,n}$ described so far have already appeared in the proof of Theorem 2.6.) Finally, suppose that $r = 2k + 1$ and $n = 2\ell + 1$ are both odd. In this case, $H_{r,n}$ is obtained from $H_{r-1,n}$ by adding the edge $v_i v_{i+\ell}$ for each i ($1 \leq i \leq n$) and the edge $v_1 v_{\ell+2}$. Therefore, when r and n are both odd, $H_{r,n}$ contains one vertex of degree $r + 1$ and $n - 1$ vertices of degree r . If r or n is even, then the size of $H_{r,n}$ is $\frac{rn}{2}$; while if r and n are odd, then the size of $H_{r,n}$ is $\frac{rn+1}{2}$. In general, the size of $H_{r,n}$ is $\lceil \frac{rn}{2} \rceil$. We note that if r or n is even, then the graphs $H_{r,n}$ were also encountered in Theorem 2.6. The graphs $H_{4,8}$, $H_{5,8}$, and $H_{5,9}$ are shown in Figure 5.10. The graphs $H_{r,n}$ are referred to as the **Harary graphs**, named for Frank Harary. We will visit Harary again in Chapter 6. Our interest in the Harary graphs comes from the following fact.

Theorem 5.14 *For every two integers r and n with $2 \leq r < n$,*

$$\kappa(H_{r,n}) = r.$$

Although it is tedious to verify Theorem 5.14 in general, we discuss a proof for some small values of r . Since $H_{2,n} = C_n$, it follows that $\kappa(H_{2,n}) = 2$. We now verify Theorem 5.14 for $r = 3$.

Example 5.15 *Show that $\kappa(H_{3,n}) = 3$.*

Figure 5.10: The Harary graphs $H_{4,8}$, $H_{5,8}$, and $H_{5,9}$

Solution. Suppose first that $n = 2\ell \geq 4$ is even. Since $H_{3,n}$ contains a spanning n -cycle $C : v_1, v_2, \dots, v_n, v_1$ and $H_{3,n}$ is 3-regular, it follows that

$$2 \leq \kappa(H_{3,n}) \leq 3.$$

Let $U = \{u, v\}$ be a set of two vertices of $H_{3,n}$. We show that $H_{3,n} - U$ is connected. There is a $u - v$ path P on C of length at most ℓ . Without loss of generality, we may assume that $u = v_1$ and $v = v_j$, where $2 \leq j \leq \ell + 1$. If $v = v_2$, then $H_{3,n} - U$ contains the spanning path $v_3, v_4, \dots, v_n$ and so $H_{3,n} - U$ is connected. Thus $3 \leq j \leq \ell + 1$. However then, $H_{3,n} - U$ contains the paths

$$P' : v_2, v_3, \dots, v_{j-1} \text{ and } P'' : v_{j+1}, v_{j+2}, \dots, v_n$$

and the edge $v_2 v_{\ell+2}$. Since $v_{\ell+2}$ belongs to P'' , the graph $H_{3,n} - U$ is connected. Hence, when n is even, $H_{3,n}$ does not contain a vertex-cut of cardinality 2.

Next, suppose that $n = 2\ell + 1 \geq 5$ is odd. Let U be a set of two vertices, say $U = \{v_i, v_j\}$, where $1 \leq i < j \leq n$. If $j = i + 1$, then $H_{3,n} - U$ contains the spanning path $v_{j+1}, v_{j+2}, \dots, v_{i-1}$, where the subscripts are expressed modulo n and so $H_{3,n} - U$ is connected. Thus, we may assume that $j \neq i + 1$.

Either the path $P : v_i, v_{i+1}, \dots, v_j$ or the path $Q : v_j, v_{j+1}, \dots, v_i$ has length ℓ or less. If the length of P is ℓ or less, then $H_{3,n} - U$ contains the paths

$$P' : v_{i+1}, v_{i+2}, \dots, v_{j-1} \text{ and } Q' : v_{j+1}, v_{j+2}, \dots, v_{i-1}$$

as well as the edge $v_{i+1} v_{i+\ell+1}$, where $v_{i+\ell+1}$ belongs to Q' and so $H_{3,n} - U$ is connected. If the length of Q is ℓ or less, then $H_{3,n} - U$ contains the paths P' and Q' as well as the edge $v_{j+2} v_{j+\ell+2}$, where $v_{j+\ell+2}$ belongs to P' and $H_{3,n} - U$ is connected. Since U is not a cut-vertex, $\kappa(H_{3,n}) = 3$. $\diamond$

Let us return to Theorem 5.13, where we noted that if G is a graph of order n and size $m \geq n - 1$, then $\kappa(G) \leq \lfloor \frac{2m}{n} \rfloor$. For every two integers m and n such that $1 \leq n - 1 \leq m \leq \binom{n}{2}$, there exists a unique positive integer r such that $r \leq \frac{2m}{n} < r + 1$. Consequently,

$$\frac{rn}{2} \leq \left\lceil \frac{rn}{2} \right\rceil \leq m < \frac{n(r+1)}{2}.$$

Thus the smallest size of a graph G of order n such that $\kappa(G) = r$ is $\lceil \frac{rn}{2} \rceil$, which, as we have seen, is the size of the Harary graph $H_{r,n}$. Since $\lceil \frac{rn}{2} \rceil \leq m < \frac{n(r+1)}{2}$, it follows that $m = \lceil \frac{rn}{2} \rceil + t$, where $0 \leq t < n/2$ if n is even, $0 \leq t < (n-1)/2$ if r is even and n is odd, and $0 \leq t < (n-3)/2$ if r and n are both odd. Adding t edges to the graph $H_{r,n}$ produces a graph H' of size m . Since $H_{r,n}$ is a spanning subgraph of H' , it follows that

$$r = \kappa(H_{r,n}) \leq \kappa(H').$$

Necessarily, H' contains one or more vertices of degree r and so $\delta(H') = r$. It then follows by Theorem 5.11 that $\kappa(H') \leq r$. Therefore, $\kappa(H') = r$, as claimed, and the bound in Theorem 5.13 is sharp.

Exercises for Section 5.3

- 5.17 Does it make sense to define the concept of a minimal vertex-cut in a graph? If so, how would this be defined and what would be a natural question to ask? If not, why is this the case?
- 5.18 Let PG be the Petersen graph. Give an example of
- (a) a minimum vertex-cut in PG.
 - (b) a vertex-cut U in PG such that U is not a minimum vertex-cut of PG and no proper subset of U is a vertex-cut of PG.
- 5.19 Prove or disprove: Let G be a nontrivial graph. For every vertex v of G , $\kappa(G - v) = \kappa(G)$ or $\kappa(G - v) = \kappa(G) - 1$.
- 5.20 Let G be a connected graph of order $n \geq 4$ and let k be an integer with $2 \leq k \leq n - 2$.
- (a) Prove that if G is not k -connected, then G contains a vertex-cut U with $|U| = k - 1$.
 - (b) Prove that if G is not k -edge-connected, then G contains an edge-cut X with $|X| = k - 1$.
- 5.21 Give an example of a graph with the following properties or explain why no such example exists.
- (a) a 2-connected graph that is not 3-connected.
 - (b) a 3-connected graph that is not 2-connected.
 - (c) a 2-edge-connected graph that is not 3-edge-connected.
 - (d) a 3-edge-connected graph that is not 2-edge-connected.
- 5.22 (a) Prove that if G is a k -connected graph and e is an edge of G , then $G - e$ is $(k - 1)$ -connected.

- (b) Prove that if G is a k -edge-connected graph and e is an edge of G , then $G - e$ is $(k - 1)$ -edge-connected.
- 5.23 (a) Prove that if G is a k -connected graph, then $G + K_1$ is $(k + 1)$ -connected.
 (b) Prove that if G is a k -edge-connected graph, then $G + K_1$ is $(k + 1)$ -edge-connected.
- 5.24 Let G be a graph of order n and let k be an integer with $1 \leq k \leq n - 1$. Prove that if $\delta(G) \geq (n + k - 2)/2$, then G is k -connected.
- 5.25 Give an example of a graph G with the following properties or explain why no such example exists.
 - (a) $\kappa(G) = 2$, $\lambda(G) = 3$, and $\delta(G) = 4$
 - (b) $\kappa(G) = 3$, $\lambda(G) = 2$, and $\delta(G) = 4$
 - (c) $\kappa(G) = 3$, $\lambda(G) = 3$, and $\delta(G) = 2$
 - (d) $\kappa(G) = 2$, $\lambda(G) = 2$, and $\delta(G) = 3$
- 5.26 Prove that if G is a graph of order n such that $\delta(G) \geq (n - 1)/2$, then $\lambda(G) = \delta(G)$.
- 5.27 Prove or disprove:
 - (a) If G is a graph with $\kappa(G) = k \geq 1$, then $G - U$ is disconnected for every set U of k vertices of G .
 - (b) If G is a graph with $\lambda(G) = k \geq 1$, then $G - X$ is disconnected for every set X of k edges of G .
 - (c) If G is a connected graph and U is a minimum vertex-cut, then $G - U$ contains exactly two components.
 - (d) If G is a graph of order n that is not complete and contains a vertex v of degree $n - 1$, then v belongs to every vertex-cut of G .
 - (e) If G is a graph of order n that contains a vertex v of degree $n - 1$, then every edge-cut of G contains an edge incident with v .
- 5.28 (a) Show that if G is a 0-regular graph, then $\kappa(G) = \lambda(G)$.
 (b) Show that if G is a 1-regular graph, then $\kappa(G) = \lambda(G)$.
 (c) Show that if G is a 2-regular graph, then $\kappa(G) = \lambda(G)$.
 (d) By (a) – (c) and Theorem 5.12, if G is r -regular, where $0 \leq r \leq 3$, then $\kappa(G) = \lambda(G)$. Find the minimum positive integer r for which there exists an r -regular graph G such that $\kappa(G) \neq \lambda(G)$. Verify your answer.
 (e) Find the minimum positive integer r for which there exists an r -regular graph G such that $\lambda(G) \geq \kappa(G) + 2$.

- (f) The problem in (e) should suggest another question to you. Ask and answer such a question.

5.29 Give an example of

- (a) a connected graph G such that every vertex of G belongs to a minimum vertex-cut but some edge of G belongs to no minimum edge-cut.
- (b) a connected graph H such that every edge of H belongs to a minimum edge-cut but some vertex of H belongs to no minimum vertex-cut.

5.30 For a graph G , define $\bar{\kappa}(G) = \max\{\kappa(H)\}$ and $\bar{\lambda}(G) = \max\{\lambda(H)\}$, each maximized over all subgraphs H of G . How are $\bar{\kappa}(G)$ and $\bar{\lambda}(G)$ related to $\kappa(G)$ and $\lambda(G)$, respectively, and to each other?

5.31 In the graph G of Figure 5.11, the vertices represent street intersections and the edges represent roads.

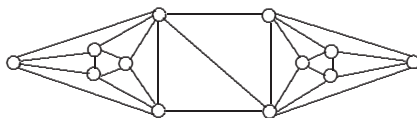


Figure 5.11: The graph in Exercise 5.31

- (a) What is the maximum number k such that if road repairs are done at the same time to any k roads (making use of these roads impossible), then it is still possible to travel between every two intersections?
 - (b) What is the maximum number k such that if intersection repairs are done at the same time to any k intersections (making use of these intersections impossible), then it is still possible to travel between every two intersections that are not under repair?
- 5.32 Verify Theorem 5.14 for $r = 4$, that is, prove that the connectivity of the Harary graph $H_{4,n}$ is 4 for every $n \geq 5$.

5.4 Menger's Theorem

We have now seen that one measure of the connectedness of a graph is its vertex-connectivity, which depends on the minimum number of vertices that must be removed to result in a disconnected or trivial graph. We will see that connectivity can be looked at in another manner.

A set S of vertices of a graph G is said to **separate** two vertices u and v of G if $G - S$ is disconnected and u and v belong to different components of $G - S$. Thus, if S separates u and v , then surely u and v are nonadjacent vertices and S is a vertex-cut of G . Certainly, the cardinality of S must be at least as large as $\kappa(G)$. Such a set S is called a $u - v$ **separating set**. A $u - v$ separating set of minimum cardinality is called a **minimum $u - v$ separating set**. An **internal vertex** of a $u - v$ path P is a vertex of P different from u and v . A collection $\{P_1, P_2, \dots, P_k\}$ of $u - v$ paths is called **internally disjoint** if every two of these paths have no vertices in common other than u and v .

There are many theorems in mathematics which state the minimum number of elements in some set equals the maximum number of elements in some other set. The following theorem is such a “min-max” theorem. It is referred to as **Menger's Theorem**.

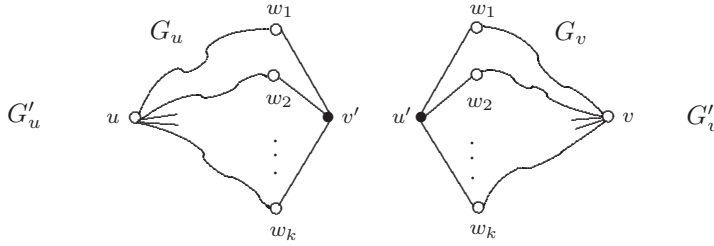
Theorem 5.16 (Menger's Theorem) *Let u and v be nonadjacent vertices in a graph G . The minimum number of vertices in a $u - v$ separating set equals the maximum number of internally disjoint $u - v$ paths in G .*

Proof. [induction] We proceed by induction on the size of the graph. Certainly, the result is true vacuously for all empty graphs. Assume that the result is true for all graphs of size less than m , where m is a positive integer, and let G be a graph of size m . Let u and v be two nonadjacent vertices of G . Suppose that there are k vertices in a minimum $u - v$ separating set. Certainly, G can contain no more than k internally disjoint $u - v$ paths. We show, in fact, that G contains k internally disjoint $u - v$ paths. We consider three cases.

Case 1. There exists a minimum $u - v$ separating set U in G containing a vertex x that is adjacent to u and v . Then the size of the subgraph $G - x$ is less than m and $U - \{x\}$ is a minimum $u - v$ separating set in $G - x$ consisting of $k - 1$ vertices. By the induction hypothesis, there are $k - 1$ internally disjoint $u - v$ paths in $G - x$. These paths together with the path u, x, v constitute k internally disjoint $u - v$ paths in G .

Case 2. There exists a minimum $u - v$ separating set W in G containing a vertex in W that is not adjacent to u and a vertex in W that is not adjacent to v . Let $W = \{w_1, w_2, \dots, w_k\}$. Let G_u be the subgraph of G consisting of all $u - w_i$ paths in G , where only $w_i \in W$ for each i ($1 \leq i \leq k$) and let G'_u be the graph obtained from G_u by adding a new vertex v' and joining v' to each vertex w_i for $1 \leq i \leq k$. Let G_v and G'_v be defined similarly, where G'_v is obtained from G_v by adding the new vertex u' . Representations of the graphs G_u , G'_u , G_v , and G'_v are shown in Figure 5.12.

Since W contains a vertex that is not adjacent to u and a vertex that is not adjacent to v , the size of each of the graphs G'_u and G'_v is less than m . Since W is a minimum $u - v$ separating set in G'_u , it follows by the induction hypothesis that G'_u contains k internally disjoint $u - v'$ paths, each consisting of a $u - w_i$ path P_i followed by the edge $w_i v'$. Similarly, there are k internally disjoint $u' - v$ paths in G'_v , each consisting of a $w_i - v$ path Q_i preceded by the edge $u' w_i$. Since W is

Figure 5.12: The graphs G_u , G'_u , G_v , and G'_v in Case 2

a $u - v$ separating set in G , the two graphs G_u and G_v have only the vertices of W in common. Therefore, the k paths obtained by following P_i by Q_i for each i ($1 \leq i \leq k$) are internally disjoint $u - v$ paths in G .

Case 3. For each minimum $u - v$ separating set S in G , either every vertex of S adjacent to u and not adjacent to v or every vertex of S is adjacent to v and not adjacent to u . Let $P : u, x, y, \dots, v$ be a $u - v$ geodesic in G and let $e = xy$. Consider the subgraph $G - e$ in G . Certainly, every minimum $u - v$ separating set in $G - e$ contains at least $k - 1$ vertices. We claim, in fact, that a minimum $u - v$ separating set in $G - e$ contains k vertices, for assume, to the contrary, that $G - e$ contains a minimum $u - v$ separating set $Z = \{z_1, z_2, \dots, z_{k-1}\}$. Then $Z \cup \{x\}$ is a minimum $u - v$ separating set in G . Since x is adjacent to u , each vertex z_i ($1 \leq i \leq k - 1$) is also adjacent to u and not adjacent to v . On the other hand, $Z \cup \{y\}$ is a minimum $u - v$ separating set in G and each z_i ($1 \leq i \leq k - 1$) is adjacent to u . This implies that y is also adjacent to u , which contradicts the fact that P is a $u - v$ geodesic. Therefore, as claimed, k is the minimum number of vertices in a $u - v$ separating set in $G - e$. By the induction hypothesis, there are k internally disjoint $u - v$ paths in $G - e$. Hence there are k internally disjoint $u - v$ paths in G as well. ■

In the graph G of Figure 5.13, $U = \{w_1, w_2, w_3\}$ is a $u - v$ separating set. Since there is no $u - v$ separating set with fewer than three vertices, U is a minimum $u - v$ separating set. By Theorem 5.16, there are three internally disjoint $u - v$ paths in G (as indicated in bold in Figure 5.13).

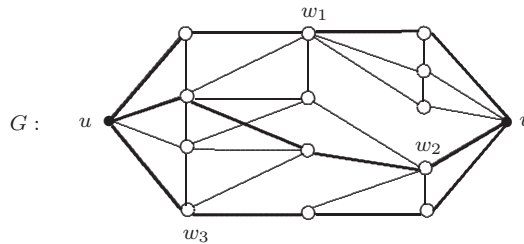


Figure 5.13: Illustrating Menger's Theorem

As we mentioned above, Theorem 5.16 is referred to as Menger's Theorem, named for Karl Menger who was born in Vienna, Austria on January 13, 1902. Menger developed a talent for mathematics and physics at an early age and entered the University of Vienna in 1920 to study physics. The following year he attended a lecture by Hans Hahn on *Neueres über den Kurvenbegriff* (*What's new concerning the concept of a curve*) and Menger's interests turned towards mathematics. In the lecture it was mentioned that there was no satisfactory definition of a curve (at that time), despite unsuccessful attempts to do so by a number of distinguished mathematicians, including Georg Cantor, Camille Jordan, and Giuseppe Peano. Some mathematicians, including Felix Hausdorff and Ludwig Bieberbach, felt that it was unlikely that this problem would ever be solved. Despite being an undergraduate with limited mathematical background, Menger solved the problem and presented his solution to Hahn. This led Menger to work on curve and dimension theory.

Menger became quite ill while a student and his studies were interrupted. During this time both of his parents died. He eventually returned to the university where he completed his studies under Hahn in 1924. The next year Menger went to Amsterdam to work with Luitzen Brouwer, and Menger broadened his mathematical interests. In 1927 Menger returned to the University of Vienna to accept the position of Chair of Geometry. It was during that year that he published the paper "Zur allgemeinen Kurventheorie" (Menger's Theorem). Menger himself referred to this result as the " n -arc theorem" and proved it as a lemma for a theorem in curve theory.

In the spring of 1930, Karl Menger traveled to Budapest and met many Hungarian mathematicians, including Dénes König. Menger had read some of König's papers. During his visit, Menger learned that König was working on a book that would contain what was known at that time about graph theory. Menger was pleased to hear this and mentioned his theorem to König, which had only been published three years earlier as a lemma in a paper on curve theory. König was not aware of Menger's work and, in fact, didn't believe that the theorem was true. Indeed, the very evening of their meeting, König set out to construct a counterexample. When the two met again the next day, König greeted Menger with "A sleepless night!". König then asked Menger to describe his proof, which he did. After that, König said that he would add a final section to his book on the theorem, which he did. This was a major reason why Menger's Theorem became so widely known among those interested in graph theory.

Because of the political situation in Austria in 1938, Menger left for a position in the United States at the University of Notre Dame. While there, his work emphasized geometry, which didn't have the impact of his earlier work since geometry was not a subject of great interest to many, especially in the United States. Menger went to the Illinois Institute of Technology in 1948 and spent the rest of his life in the Chicago area. Menger was considered one of the leading mathematicians of the 20th century. He died on October 5, 1985.

With the aid of Menger's Theorem, Hassler Whitney was able to present a characterization of k -connected graphs.

Theorem 5.17 *A nontrivial graph G is k -connected for some integer $k \geq 2$ if and only if for each pair u, v of distinct vertices of G there are at least k internally disjoint $u - v$ paths in G .*

Proof. [direct proof] Assume that G is a k -connected graph, where $k \geq 2$. Let u and v be two distinct vertices of G . Suppose first that u and v are not adjacent and let U be a minimum $u - v$ separating set. Then $|U| \geq \kappa(G) \geq k$. By Theorem 5.16, G contains at least k internally disjoint $u - v$ paths. Next, suppose that u and v are adjacent, where $e = uv$. Then $G - e$ is $(k - 1)$ -connected (see Exercise 5.22). Let W be a minimum $u - v$ separating set in $G - e$. Thus

$$|W| \geq \kappa(G - e) \geq k - 1.$$

By Theorem 5.16, $G - e$ contains at least $k - 1$ internally disjoint $u - v$ paths, and so G contains at least k internally disjoint $u - v$ paths.

For the converse, assume that G is a graph containing at least k internally disjoint $u - v$ paths for every pair u, v of distinct vertices of G . Let U be a minimum vertex-cut of G . Then $|U| = \kappa(G)$. Let x and y be vertices in distinct components of $G - U$. Thus U is an $x - y$ separating set of G . Since there are at least k internally disjoint $x - y$ paths in G , it follows by Theorem 5.16 that $\kappa(G) = |U| \geq k$. Therefore, G is k -connected. ■

While Whitney's Theorem (Theorem 5.17) is a valuable consequence of Menger's Theorem, the next result (Corollary 5.18) follows from Whitney's Theorem. Corollary 5.19 is, in turn, a consequence of Corollary 5.18. Proofs of both corollaries are left as exercises.

Corollary 5.18 *Let G be a k -connected graph and let S be any set of k vertices. If a graph H is obtained from G by adding a new vertex w and joining w to the vertices of S , then H is also k -connected.*

We have mentioned that a collection $\{P_1, P_2, \dots, P_k\}$ of $u - v$ paths are internally disjoint if every two distinct paths in the collection have only u and v in common, that is, each internal vertex in one of the paths lies on no other path in the collection. More generally, for $k + 1$ distinct vertices $u, v_1, v_2, \dots, v_k$, a collection $\{P_1, P_2, \dots, P_k\}$ of k paths, where P_i is a $u - v_i$ path ($1 \leq i \leq k$), are **internally disjoint** if every two distinct paths in the collection have only u in common. For $2k$ distinct vertices $u_1, u_2, \dots, u_k, v_1, v_2, \dots, v_k$, a collection $\{P_1, P_2, \dots, P_k\}$ of k paths, where P_i is a $u_i - v_i$ path ($1 \leq i \leq k$), are **disjoint** if no two distinct paths in the collection have a vertex in common.

Corollary 5.19 *If G is a k -connected graph and $u, v_1, v_2, \dots, v_k$ are $k + 1$ distinct vertices of G , then there exist internally disjoint $u - v_i$ paths ($1 \leq i \leq k$) in G .*

We saw in Theorem 5.7 that if a graph G is a 2-connected graph (that is, a nonseparable graph of order at least 3), then every two vertices of G lie on a common cycle. Gabriel Dirac obtained a generalization of this result for k -connected graphs.

Theorem 5.20 *If G is a k -connected graph, $k \geq 2$, then every k vertices of G lie on a common cycle of G .*

Proof. [proof by contradiction] Let $S = \{v_1, v_2, \dots, v_k\}$ be a set of k vertices of G . We show that there exists a cycle in G containing every vertex of S . Among all cycles in G , let C be one containing a maximum number ℓ of vertices of S . We claim that $\ell = k$. Assume, to the contrary, that $\ell < k$. Since G is k -connected, $k \geq 2$, it follows that G is 2-connected and so $2 \leq \ell < k$ by Theorem 5.7. We may assume that C contains the vertices $v_1, v_2, \dots, v_\ell$ of S and that the vertices of S on C appear in the order $v_1, v_2, \dots, v_\ell$ as we proceed cyclically about C . In particular, there is a $v_1 - v_2$ path P on C that contains no internal vertices belonging to S . Since $\ell < k$, there is a vertex $u \in S$ that does not belong to C . Furthermore, since $2 \leq \ell < k$, the graph G is ℓ -connected as well. Applying Corollary 5.19 to the vertices $u, v_1, v_2, \dots, v_\ell$, we see that G contains internally disjoint $u - v_i$ paths P_i ($1 \leq i \leq \ell$). Replacing P by P_1 and P_2 produces a cycle C' containing the vertices $u, v_1, v_2, \dots, v_\ell$, a contradiction. ■

All of the results mentioned in this section concern connectivity. There are also edge-connectivity analogues of both Theorems 5.16 and 5.17. We state these next.

Theorem 5.21 *For distinct vertices u and v in a graph G , the minimum number of edges of G that separate u and v equals the maximum number of pairwise edge-disjoint $u - v$ paths in G .*

Theorem 5.22 *A nontrivial graph G is k -edge-connected if and only if G contains k pairwise edge-disjoint $u - v$ paths for each pair u, v of distinct vertices of G .*

Exercises for Section 5.4

- 5.33 Let G be a 5-connected graph and let u, v , and w be three distinct vertices of G . Prove that G contains two cycles C and C' that have only u and v in common but neither of which contains w .
- 5.34 Prove Corollary 5.18: Let G be a k -connected graph and let S be any set of k vertices. If a graph H is obtained from G by adding a new vertex w and joining w to the vertices of S , then H is also k -connected.
- 5.35 Prove Corollary 5.19: If G is a k -connected graph and $u, v_1, v_2, \dots, v_k$ are $k+1$ distinct vertices of G , then there exist internally disjoint $u - v_i$ paths ($1 \leq i \leq k$) in G .
- 5.36 Let G be a k -connected graph of order $n \geq 2k$ and let U and W be two disjoint sets of k vertices of G . Prove that there exist k disjoint paths connecting U and W .
- 5.37 Determine the connectivity and edge-connectivity of the n -cube Q_n .

5.5 Exploration: Geodetic Sets

By Theorem 5.17, a graph G is k -connected if and only if there are at least k internally disjoint $u - v$ paths for each pair u, v of distinct vertices of G . That is, the connectivity of a graph G can also be measured in terms of the number of internally disjoint $u - v$ paths for all pairs u, v of distinct vertices of G . There is no restriction on the length of these paths except, of course, the length of each path must be at least $d(u, v)$.

Recall that a $u - v$ path of length $d(u, v)$ is called a $u - v$ geodesic. Rather than considering the maximum number of internally disjoint $u - v$ paths for non-adjacent vertices u and v , we now change our emphasis from internally disjoint $u - v$ paths to $u - v$ geodesics for pairs u, v of distinct (not necessarily nonadjacent) vertices. For two vertices u and v in a connected graph G , the **closed interval** $I[u, v]$ of u and v is the set of those vertices belonging to at least one $u - v$ geodesic. Necessarily, $u, v \in I[u, v]$ for every two distinct vertices u and v ; in fact, if $uv \in E(G)$, then $I[u, v] = \{u, v\}$. For a set S of vertices in a connected graph G , the **closed interval** of S is the set

$$I[S] = \bigcup_{u, v \in S} I[u, v].$$

For example, in the graph G of Figure 5.14, $d(u_1, w_3) = 3$ and $I[u_1, w_3] = V(G) - \{w_1, v_2, u_3\}$, that is, every vertex of G lies on some $u_1 - w_3$ geodesic except w_1 , v_2 , and u_3 . On the other hand, $d(w_1, u_3) = 4$ and $I[w_1, u_3] = V(G) - \{v_2\}$. Actually, there do not exist two vertices x and y such that $I[x, y] = V(G)$. This brings up the question of finding sets $S \subseteq V(G)$ for which $I[S] = V(G)$.

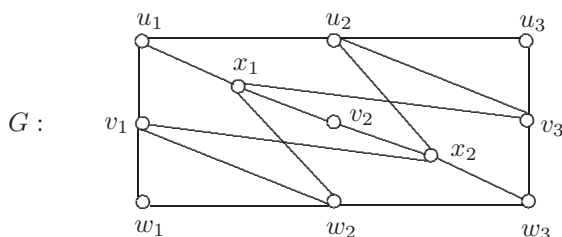


Figure 5.14: Closed intervals in a graph

Let G be a connected graph. A set S of vertices in G is called a **geodetic set** for G if $I[S] = V(G)$. A geodetic set of minimum cardinality in G is called a **minimum geodetic set** of G and this cardinality is the **geodetic number** of G , which is denoted by $g(G)$. For the graph G of Figure 5.14, the set $S = \{w_1, v_2, u_3\}$ is a geodetic set since $I[S] = V(G)$. Because no geodetic set in G consists of two vertices, S is a minimum geodetic set and so $g(G) = |S| = 3$.

A vertex v in a graph G is called an **extreme vertex** (or a **complete** or **simplicial vertex**) if the subgraph induced by its neighborhood is complete. For

example, the vertex w_1 and u_3 are extreme vertices of the graph G of Figure 5.14. In particular, every end-vertex in a graph is extreme.

Theorem 5.23 *Every geodetic set of a connected graph G contains its extreme vertices. In particular, if the set S of extreme vertices is a geodetic set of G , then S is the unique minimum geodetic set of G .*

Proof. [proof by contradiction] Let x be an extreme vertex of a connected graph G . Suppose that x lies on a $u - v$ geodesic P . We show that $x = v$ or $x = u$. Assume, to the contrary, that x is an internal vertex of P . Then x is adjacent to two nonadjacent vertices of P , contradicting the fact that x is an extreme vertex. Thus, every geodetic set of G contains its extreme vertices. Therefore, if the set S of extreme vertices is a geodetic set of G , then S is the unique minimum geodetic set of G . ■

Exercises for Section 5.6

- 5.38 Determine the geodetic number of K_n for $n \geq 2$.
- 5.39 Determine the geodetic number of C_n for $n \geq 3$.
- 5.40 Determine the geodetic number of $K_{s,t}$ for $1 \leq s \leq t$.
- 5.41 Determine the geodetic number of Q_n for $n \geq 1$.
- 5.42 (a) Let T be a tree with k end-vertices. Prove that $g(T) = k$.
 (b) For each pair k, n of positive integers with $2 \leq k \leq n$, find a connected graph G of order n with $g(G) = k$.
- 5.43 Let S be a minimum geodetic set in a nontrivial connected graph G . Prove that no cut-vertex in G belongs to S .
- 5.44 It was mentioned that $S = \{w_1, v_2, u_3\}$ is a minimum geodetic set for the graph G of Figure 5.14. Is there any other minimum geodetic set for this graph?
- 5.45 Does there exist a connected graph G with two or more minimum geodetic sets and a vertex v that is not an extreme vertex of G such that v belongs to every minimum geodetic set of G ?
- 5.46 Does there exist a connected graph G with a unique minimum geodetic set S , where no vertex of S is an extreme vertex of G ? If the answer is yes, then give an example of such a graph with a large geodetic number.
- 5.47 (a) Give an example of a connected graph G with $g(G) = 3$ containing two disjoint minimum geodetic sets.

- (b) The problem in (a) should suggest another problem to you. Propose and solve such a problem.

5.48 For a set S of vertices of a connected graph G , define the **total distance** $td(S)$ of S by

$$td(S) = \sum_{u,v \in S} d(u,v).$$

- (a) Does there exist a connected graph G containing distinct minimum geodetic sets S_1 and S_2 such that $td(S_1) \neq td(S_2)$?
- (b) For each integer $k \geq 2$, give an example of a connected graph G with $g(G) = k$ and a minimum geodetic set S such that $d(u,v) = k$ for every two distinct vertices u and v in S .
- (c) Does your example in (b) suggest another question?

Chapter 6

Traversability

6.1 Eulerian Graphs

Figure 6.1(a) shows the layout of a housing development in a community, where mailboxes are placed along one side of each street (indicated by double lines in the diagram). Can a letter carrier make a round trip through the development and pass by each mailbox but once? Figure 6.1(b) shows that the answer to this question is yes.

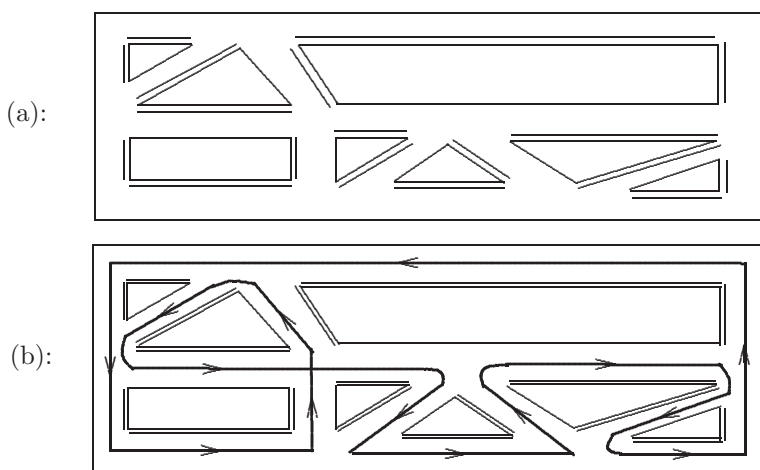


Figure 6.1: A housing development and a route of a letter carrier

The streets in the housing development can be represented quite naturally by the graph G of Figure 6.2, where the vertices represent the street intersections and the edges represent the streets. In terms of graph theory, the question for

and the other Bernoullis were numerous, one of Johann's major accomplishments may have been to convince the father of a young Leonhard Euler (pronounced OY-ler) to have his son discontinue studying theology and study mathematics instead. Later Johann Bernoulli became the mathematical advisor of Euler. When one individual serves as the academic advisor (usually doctoral advisor) of another, the advisor is referred by some as the **academic father** (or **academic mother**) of the student. This provides a sense of "family" for teacher and student. Hence Johann Bernoulli could be called the academic father of Euler.

Euler was born in Basel, Switzerland on April 15, 1707. While in his 20s, he became ill and lost vision in one of his eyes. Later, he developed a cataract in his other eye and spent the last few years of his life totally blind. However, just as the magnificent composer Ludwig van Beethoven did much of his work while totally deaf, Euler did much of his mathematical research while totally blind. During his lifetime, more than 500 research papers and books of his were published. After his death (from a stroke) on September 18, 1783, another 400 were published. At that time and for a good many years afterwards, Euler had more publications than any other mathematician, only to be exceeded in the 20th century by Paul Erdős (pronounced AIR-dish), an academic descendant of Euler. We will visit Paul Erdős later.

While Euler made significant contributions to every area of mathematics that existed during his lifetime, it is a contribution he made to an area of mathematics that did not exist during his lifetime that is of primary interest to us here.

The city of Königsberg (located in Northern Germany) would play an interesting role in Euler's life and in the history of graph theory. The River Pregel flowed through Königsberg, separating it into four land areas. Seven bridges were built over the river that allowed the citizens of Königsberg to travel between these land areas. A map of Königsberg, showing the four land areas (labeled A, B, C, D), the location of the river, and the seven bridges at that time are given in Figure 6.4.

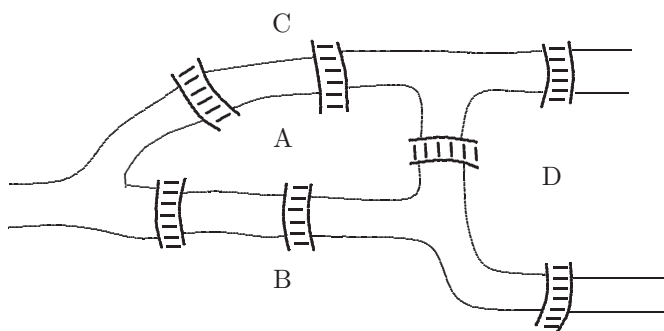


Figure 6.4: Königsberg in the early 18th century

The story goes that the citizens of Königsberg enjoyed going for walks near the river. Some citizens wondered whether it was possible to go for a walk in

Königsberg and pass over each bridge exactly once. This became known as the **Königsberg Bridge Problem**. Evidently this problem remained unsolved for some time and became well-known throughout the region. This problem eventually came to the attention of Euler (who was believed to be in St. Petersburg at the time). Euler observed that this situation could be represented by the diagram (actually the multigraph) shown in Figure 6.5, where the vertices represent the four land areas and the edges represent the seven bridges. (The subject of graph theory did not exist before this time but it was about to be born.)

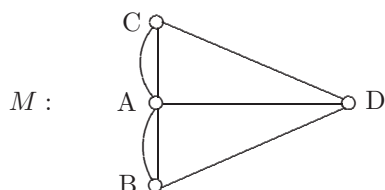


Figure 6.5: The Königsberg multigraph

When considering a walk W in some multigraph M , it is not enough to express W as a sequence of vertices. For example, if u, v are consecutive vertices in W and there are parallel edges joining u and v in M , then it must be indicated somehow which edge is being traversed, such as u, e, v .

In graph theory terms, the Königsberg Bridge Problem is: Does the multigraph M of Figure 6.5 contain an Eulerian circuit or an Eulerian trail? (Of course, Euler didn't use these terms.) Suppose that such a journey over the seven bridges of Königsberg was possible. Then it must begin at some land area and end at a land area (possibly the same one). Such a journey must therefore result in a trail of length 7 (one edge for each bridge) that encounters eight vertices (including repetition), namely,

$$T : v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8.$$

Each of the vertices v_i ($1 \leq i \leq 8$) represents a land area (A, B, C, or D). Certainly, each land area must appear in T . Observe that all four vertices of the multigraph M have odd degree. At least two vertices of M are neither the initial nor the terminal vertex of T . Thus, each time that such a vertex is entered along T , it is exited. However, this implies that the vertex has even degree, which is impossible. Consequently, a walk that passes over every bridge in Königsberg exactly once is impossible as well.

Königsberg was founded in 1255 and was the capital of German East Prussia. The Prussian Royal Castle was located in Königsberg, but it was destroyed during World War II, as was much of the city. Because of the outcome of the war, it was decided at the Potsdam Conference in 1945 that a region located between Poland and Lithuania, containing the city of Königsberg, should be made part of Russia. In 1946, Königsberg was renamed Kaliningrad after Mikhail Kalinin, the formal leader of the Soviet Union during 1919–1946. After the fall of the

USSR, Lithuania and other former Soviet republics became independent, and Kaliningrad was no longer part of Russia. None of the attempts to change its name back to Königsberg have been successful.

Returning to the multigraph M of Figure 6.5, we are led to ask what characteristic it possesses that doesn't allow it to have an Eulerian circuit or an Eulerian trail. We have already mentioned that all four vertices of M have odd degree. This, it turns out, is the key observation. For a connected graph G to be Eulerian, it is both necessary and sufficient that each vertex of G has even degree.

Theorem 6.1 *A nontrivial connected graph G is Eulerian if and only if every vertex of G has even degree.*

Proof. [direct proof, proof by contradiction] Assume first that G is Eulerian. Then G contains an Eulerian circuit C . Suppose that C begins at the vertex u (and therefore ends at u). We show that every vertex of G is even. Let v be a vertex of G different from u . Since C neither begins nor ends at v , each time that v is encountered on C , two edges are accounted for (one to enter v and another to exit v). Thus v has even degree. Now to u . Since C begins at u , this accounts for one edge. Another edge is accounted for because C ends at u . If u is encountered at other times, two edges are accounted for. So u is even as well.

For the converse, assume that G is a nontrivial connected graph in which every vertex is even. We show that G contains an Eulerian circuit. Among all trails in G , let T be one of maximum length. Suppose that T is a $u - v$ trail. We claim that $u = v$. If not, then T ends at v . It is possible that v may have been encountered earlier in T . Each such encounter involves two edges of G , one to enter v and another to exit v . Since T ends at v , an odd number of edges at v have been encountered. But v has even degree. This means that there is at least one edge at v , say vw , that does not appear on T . But then T can be extended to w , contradicting the assumption that T has maximum length. Thus T is a $u - u$ trail, that is, $C = T$ is a $u - u$ circuit. If C contains all edges of G , then C is an Eulerian circuit and the proof is complete.

Suppose then that C does not contain all edges of G , that is, there are some edges of G that do not lie on C . Since G is connected, some edge $e = xy$ not on C is incident with a vertex x that is on C . Let $H = G - E(C)$, that is, H is the spanning subgraph of G obtained by deleting the edges of C . Every vertex of C is incident with an even number of edges on C . Since every vertex of G has even degree, every vertex of H has even degree. It is possible, however, that H is disconnected. On the other hand, H has at least one nontrivial component, namely, the component H_1 of H containing the edge xy . This means that H_1 is connected and every vertex of H_1 has even degree. Consider a trail of maximum length in H_1 , beginning at x . As we just saw, this trail must also end at x and is an $x - x$ circuit C' of H_1 .

Now if, in the circuit C , we were to attach C' when we arrive at x , we obtain a circuit C'' in G of greater length than C , which is a contradiction. This implies that C contains all edges of G and is an Eulerian circuit. ■

Although Theorem 6.1 is credited to Leonhard Euler, as indeed it should, Euler's proof was incomplete. Euler failed to show that if every vertex of a connected graph G is even, then G is Eulerian. While even in those days when mathematicians often discussed a proof rather than wrote it with the care and precision that is normally done today, it is quite likely that Euler felt this implication was clear. Nevertheless, it wasn't until 1873 that the missing portion of the proof was published. The proof was completed by Carl Hierholzer. Actually at that time Hierholzer had died but he had told a colleague what he had done, and his colleague kindly wrote the paper for him and listed Hierholzer as its sole author.

With the aid of Theorem 6.1, it is now easy to characterize graphs possessing an Eulerian trail.

Corollary 6.2 *A connected graph G contains an Eulerian trail if and only if exactly two vertices of G have odd degree. Furthermore, each Eulerian trail of G begins at one of these odd vertices and ends at the other.*

Proof. [direct proof] Assume first that G contains an Eulerian trail T . Thus T is a $u - v$ trail for some distinct vertices u and v . We now construct a new connected graph H from G by adding a new vertex x of degree 2 and joining it to u and v . Then $C : T, x, u$ is an Eulerian circuit in H . By Theorem 6.1, every vertex of H is even and so only u and v have odd degrees in $G = H - x$.

For the converse, we proceed in a similar manner. Let G be a connected graph containing exactly two vertices u and v of odd degree. We show that G contains an Eulerian trail T , where T is either a $u - v$ trail or a $v - u$ trail. Add a new vertex x of degree 2 to G and join it to u and v , calling the resulting graph H . Therefore, H is a connected graph all of whose vertices are even. By Theorem 6.1, H is an Eulerian graph containing an Eulerian circuit C . Since it is irrelevant which vertex of C is the initial (and terminal) vertex, we assume that C is an $x - x$ circuit. Since x is incident only with the edges ux and vx , one of these is the first edge of C and the other is the final edge of C . Deleting x from C results in an Eulerian trail T of G that begins either at u or v and ends at the other. ■

As a result of Theorem 6.1 and Corollary 6.2, it is now relatively easy to determine whether a graph contains an Eulerian circuit or an Eulerian trail. Furthermore, Theorem 6.1 and Corollary 6.2 both hold if "graph" is replaced by "multigraph".

Certainly, C_n , $n \geq 3$, is Eulerian and P_n , $n \geq 2$, contains an Eulerian trail. A complete graph is Eulerian if and only if $n \geq 3$ and n is odd; while $K_{s,t}$ is Eulerian if and only if s and t are both even. The graph $K_{2,t}$ contains an Eulerian trail if and only if t is odd. The n -cube Q_n is Eulerian if and only if $n \geq 2$ and n is even. We have seen that the n -cube Q_n , $n \geq 2$, is defined as the Cartesian product of the $(n-1)$ -cube Q_{n-1} and K_2 . This suggests a question.

Example 6.3 Find a necessary and sufficient condition for the Cartesian product $G \times H$ of two nontrivial connected graphs G and H to be Eulerian.

Solution. We can think of $G \times H$ as being constructed by replacing each vertex v of G by a copy H_v of H . Let x be a vertex of $G \times H$. Then x belongs to H_v for some vertex v of G . The vertex x is adjacent to its neighbors in H_v as well as to one vertex in H_u for every neighbor u of v in G . Thus

$$\deg_{G \times H} x = \deg_{H_v} x + \deg_G v.$$

Hence $\deg_{G \times H} x$ is even if and only if $\deg_{H_v} x$ and $\deg_G v$ are both even or both odd (that is, they are of the same parity).

If $\deg_{H_v} x$ is even, then $\deg_G v$ is even for every vertex v of G ; while if $\deg_{H_v} x$ is odd, then $\deg_G v$ is odd for every vertex v of G . We have therefore arrived at the following:

Let G and H be nontrivial connected graphs. Then $G \times H$ is Eulerian if and only if both G and H are Eulerian or every vertex of G and H is odd. $\diamond$

Exercises for Section 6.1

- 6.1 The diagram of Figure 6.6 shows the nine rooms on the second floor of a large house with doorways between various rooms. Is it possible to start in some room and go for a walk so that each doorway is passed through exactly once? How is this question related to graph theory? Explain.

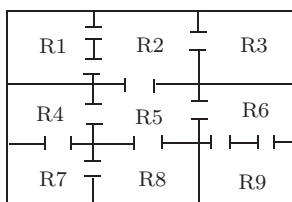


Figure 6.6: The diagram in Exercise 6.1

- 6.2 Let G_1 and G_2 be two Eulerian graphs with no vertex in common. Let $v_1 \in V(G_1)$ and $v_2 \in V(G_2)$. Let G be the graph obtained from $G_1 \cup G_2$ by adding the edge v_1v_2 . What can be said about G ?
- 6.3 Let G_1 , G_2 , and G_3 be pairwise disjoint connected regular graphs and let $G = G_1 + (G_2 + G_3)$ be the graph obtained from G_1 , G_2 , and G_3 by adding edges between every two vertices belonging to two of G_1 , G_2 , and G_3 . Prove that if G_1 and $\overline{G_1}$ are Eulerian, but G_2 and G_3 are not Eulerian, then G is Eulerian.

- 6.4 Give an example of a graph G such that
- (a) both G and $\overline{G}$ are Eulerian.
 - (b) G is Eulerian, but $\overline{G}$ is not.
 - (c) neither G nor $\overline{G}$ is Eulerian and both G and $\overline{G}$ contain an Eulerian trail.
 - (d) neither G nor $\overline{G}$ is Eulerian, but G contains an Eulerian trail and $\overline{G}$ does not.
 - (e) G contains an Eulerian trail and an edge e such that $G - e$ is Eulerian.
- 6.5 Only one graph of order 5 has the property that the addition of any edge produces an Eulerian graph. What is it?
- 6.6 Let G be a connected regular graph that is not Eulerian. Prove that if $\overline{G}$ is connected, then $\overline{G}$ is Eulerian.
- 6.7 Let G be an r -regular graph of odd order n and let $F \cong \overline{G}$, where F and G have disjoint vertex sets. A graph H is constructed from G and F by adding two new vertices u and v and joining u and v to each other as well as to every vertex of G and F . Which of the following is true? Explain.
- (a) H is Eulerian.
 - (b) H has an Eulerian trail.
 - (c) H has neither an Eulerian circuit nor an Eulerian trail.
- 6.8 (a) Show that every nontrivial connected graph G has a closed spanning walk that contains every edge of G exactly twice.
- (b) Which nontrivial connected graphs G have closed spanning walks that contain every edge of G exactly three times?

6.2 Hamiltonian Graphs

Figure 6.7 shows a diagram of a modern art museum that is divided into 15 exhibition rooms. At the end of each day, a security officer enters the reception room by the front door and checks each exhibition room to make certain that everything is in order. It would be most efficient if the officer could visit each room only once and return to the reception room. Can this be done?

This question can be rephrased in terms of graphs. A graph G can be associated with this museum where the vertices of G are the exhibition rooms and two vertices (rooms) are joined by an edge if there is a doorway between the two rooms. This graph G is shown in Figure 6.8. The question above can now be asked as follows: Does the graph G of Figure 6.8 have a cycle that contains

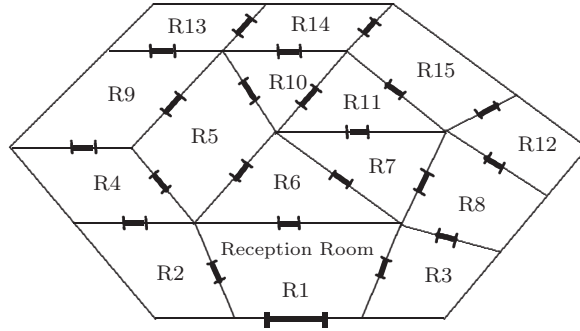


Figure 6.7: A diagram of the exhibition rooms in a museum

every vertex of G ? The answer is yes; indeed,

$C : R1, R2, R4, R9, R13, R14, R10, R5, R6, R7, R11, R15, R12, R8, R3, R1$

is such a cycle. This brings us to the main topic of this section.

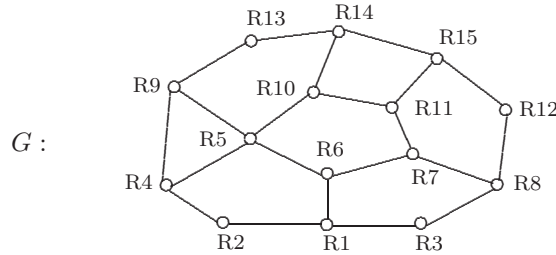


Figure 6.8: The graph that models the exhibition rooms and doorways of the museum of Figure 6.7

A cycle in a graph G that contains every vertex of G is called a **Hamiltonian cycle** of G . Thus a Hamiltonian cycle of G is a spanning cycle of G . A **Hamiltonian graph** is a graph that contains a Hamiltonian cycle. Therefore, the graph G of Figure 6.8 is Hamiltonian. Certainly the graph C_n ($n \geq 3$) is Hamiltonian. Also, for $n \geq 3$, the complete graph K_n is a Hamiltonian graph.

A path in a graph G that contains every vertex of G is called a **Hamiltonian path** in G . If a graph contains a Hamiltonian cycle, then it contains a Hamiltonian path. In fact, removing any edge from a Hamiltonian cycle produces a Hamiltonian path. If a graph contains a Hamiltonian path, however, it need not contain a Hamiltonian cycle. For example, the graph P_n clearly contains a Hamiltonian path, but P_n contains no cycles at all.

The graph $G \cong K_{3,3}$ of Figure 6.9 is a Hamiltonian graph. For example, $C : u, x, v, y, w, z, u$ is a Hamiltonian cycle of G . Since G is 3-regular and every Hamiltonian cycle contains two edges incident with each vertex of G , every

Hamiltonian cycle of the graph G of Figure 6.9 fails to contain exactly one of the three edges incident with each vertex of G . By redrawing G in Figure 6.9, we can see more easily that G is Hamiltonian. Indeed, a Hamiltonian graph of order n consists of a cycle C of length n , with possibly some additional edges joining non-consecutive vertices of C . Since a Hamiltonian cycle C in a graph G of order $n \geq 3$ is a connected 2-regular subgraph of order n , every proper subgraph of C is a path or a (disjoint) union of paths. In particular, C contains no cycle of order less than n as a subgraph, and certainly C contains no subgraph having a vertex of degree 3 or more. Furthermore, since a Hamiltonian cycle C of G is a 2-regular subgraph, if G contains a vertex v of degree 2 in G , then both edges of G incident with v must lie on C .

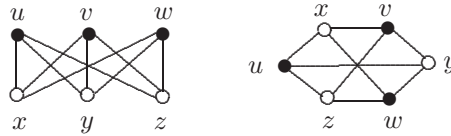


Figure 6.9: The Hamiltonian graph $K_{3,3}$

The graph G of Figure 6.10 is *not* a Hamiltonian graph. In order to see this, let's suppose that G is Hamiltonian. Then G contains a Hamiltonian cycle C . Since C contains the vertex t , which has degree 2, both tu and tz lie on C . By the same reasoning, xy and xz lie on C , as do vw and vz . However, this says that z is incident with three edges on C , which is impossible. Therefore, as we claimed, the graph G of Figure 6.10 is not Hamiltonian.

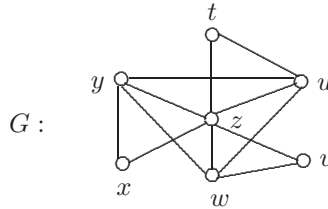


Figure 6.10: A non-Hamiltonian graph

One of the most famous non-Hamiltonian graphs is the Petersen graph (shown in Figure 6.11). The Petersen graph is a 3-regular graph of order 10. This graph can be considered as being constructed from two 5-cycles, namely the outer-cycle $C' : u_1, u_2, u_3, u_4, u_5, u_1$, the inner-cycle $C'' : v_1, v_3, v_5, v_2, v_4, v_1$, and then adding the five edges $u_1v_1, u_2v_2, u_3v_3, u_4v_4, u_5v_5$. (By interchanging the labels of C' and C'' , we see that it doesn't matter which 5-cycle is referred to as the outer-cycle or inner-cycle.) In fact, the length of a smallest cycle in the Petersen graph is 5. There are several ways to show that the Petersen graph is not Hamiltonian. We describe one of these.

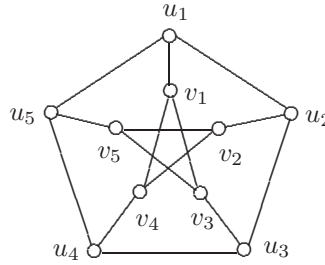
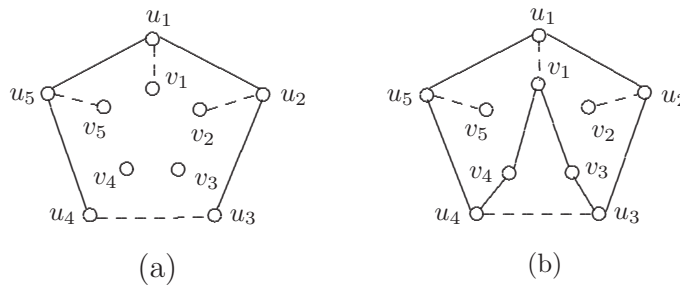


Figure 6.11: The Petersen graph: A non-Hamiltonian graph

Theorem 6.4 *The Petersen graph is non-Hamiltonian.*

Proof. [proof by contradiction] Suppose that the Petersen graph, which we will denote by PG, is Hamiltonian. Then PG contains a Hamiltonian cycle C . This cycle contains ten edges. Two of the three edges incident with each vertex of PG necessarily belong to C . Certainly, C contains all five, some, or none of the edges $u_i v_i$ ($1 \leq i \leq 5$); so at least five edges of C belong either to C' or to C'' . Therefore, either C' contains at least three edges of C , or C'' contains at least three edges of C . Without loss of generality, assume that C' contains at least three edges of C . First, observe that all five edges of C' cannot belong to C since no cycle contains a smaller cycle as a subgraph. Suppose that C contains exactly four edges of C' , say the edges $u_4 u_5, u_5 u_1, u_1 u_2, u_2 u_3$ (see Figure 6.12(a), where the dashed edges of PG cannot belong to C). However, the cycle C must then contain the edges $u_4 v_4, u_3 v_3$ as well as $v_1 v_3, v_1 v_4$ (see Figure 6.12(b)). But this implies that C contains an 8-cycle, which is impossible.

Figure 6.12: The cycle C contains exactly four edges of C'

One case remains then, namely that C contains exactly three edges of C' . There are two possibilities: (1) the three edges of C' on C are consecutive on C' and (2) these three edges are not consecutive on C' . These possibilities are shown in Figures 6.13(a) and 6.13(b), respectively. The situation in Figure 6.13(a) is impossible as $u_1 v_1$ is the only edge incident with u_1 that could lie on C . Likewise, the situation in Figure 6.13(b) is impossible since C would have to contain the

smaller cycle $u_4, v_4, v_1, v_3, u_3, u_4$. Therefore, as claimed, the Petersen graph is not Hamiltonian. ■

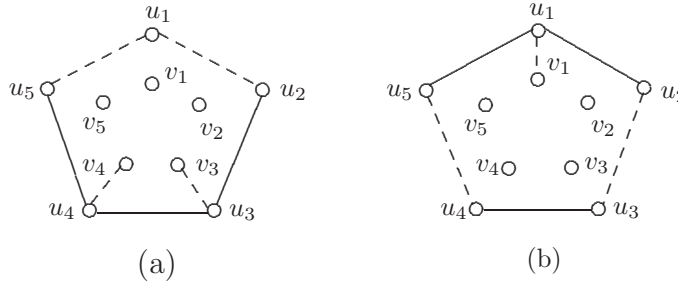


Figure 6.13: The cycle C contains exactly three edges of C'

The adjective “Hamiltonian” is named for the Irish mathematician William Rowan Hamilton (1805–1865). Although Hamilton’s personal life was filled with sorrow, his professional life was marked by numerous accomplishments. His interests included poetry, optics, astronomy, and mathematics (especially algebra). Indeed, as with poetry, Hamilton felt that mathematics too was artistic. Hamilton became acquainted with the distinguished poet William Wordsworth and the two had discussions about science and poetry. Wordsworth told Hamilton that his talents were with science, however, not poetry. One thing that worked against Hamilton, however, was his inability to write well. Hamilton was knighted in 1835 and thus became *Sir* William Rowan Hamilton. It was Hamilton who discovered quaternions (the first noncommutative algebra) and he spent much of his later years working on this topic.

Hamilton’s connection with graph theory was even more slight than Euler’s. This connection involved a well-known geometric figure. A regular dodecahedron is a polyhedron having twelve faces (all regular pentagons) and twenty corners with three edges of pentagons meeting at each corner (see Figure 6.14).

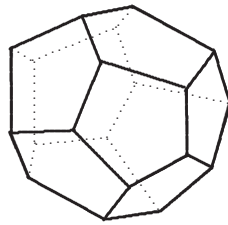


Figure 6.14: A regular dodecahedron

A regular dodecahedron has been known to be used as a desk calendar since a month can be displayed on each of its twelve faces. In 1857, Hamilton invented a game called the **Icosian Game**. It consisted of a regular dodecahedron made

from wood. Each corner of the dodecahedron was marked with a famous city of the time. The object of the game was to travel “Around the World”, that is, to find a round trip that passed through each city exactly once. To make it easier to remember which city had already been visited, there was a peg at each corner so the journey could be described by a string that was wrapped around the peg at each city visited. This game proved to be awkward and a board game version of it was produced (see Figure 6.15).

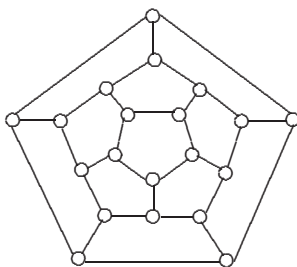


Figure 6.15: The graph of the dodecahedron

Hamilton sold the game to J. Jacques and Sons (well-known even today as makers of quality chess sets) for 25 pounds and obtained a patent for it in London in 1859. The game was a commercial failure. In graph theory terms, the object of the game was to find a Hamiltonian cycle in the graph (of the dodecahedron) in Figure 6.15. Of course, this term did not exist at that time but it was Hamilton’s game that gave rise to his name being used for these terms. Curiously, some two years before this, Thomas Penyngton Kirkman had asked in a paper of his whether there exists a cycle passing through every vertex of certain polyhedra. So Hamilton was not the first to deal with Hamiltonian graphs.

Let’s see what we know about Hamiltonian graphs. If G is a Hamiltonian graph, then certainly G is connected. Since G contains a Hamiltonian cycle, G has no cut-vertices and has order at least 3; so G is 2-connected. This says that for every vertex v of G , the graph $G - v$ is connected. If we remove more vertices from G , then the number of components of the resulting graph cannot be too large. Let $k(G)$ denote the number of components in a graph G .

Theorem 6.5 *If G is a Hamiltonian graph, then for every nonempty proper set S of vertices of G ,*

$$k(G - S) \leq |S|.$$

Proof. [direct proof] Let S be a nonempty proper subset of $V(G)$. Suppose that $k(G - S) = k$ and that $G_1, G_2, \dots, G_k$ are the components of $G - S$. Since G is Hamiltonian, G contains a Hamiltonian cycle C . Whenever C encounters a vertex of G_i for the last time ($1 \leq i \leq k$), the next vertex of C must belong to S . This implies that S must contain at least k vertices, that is, $k = k(G - S) \leq |S|$. ■

Theorem 6.5 gives a necessary condition for a graph to be Hamiltonian, that is, this theorem describes a property possessed by every Hamiltonian graph. The main benefit of a necessary condition lies in the contrapositive of the statement:

Let G be a graph. If $k(G - S) > |S|$ for some nonempty proper subset S of $V(G)$, then G is not Hamiltonian.

That is, Theorem 6.5 gives us a sufficient condition for a graph to be non-Hamiltonian. In particular, this gives us the obvious fact when a connected graph G contains a cut-vertex v (and where $S = \{v\}$) and $k(G - v) \geq 2$:

If a graph G contains a cut-vertex, then G cannot be Hamiltonian.

Consider the graph G of Figure 6.16, where $S = \{u, v, w\}$. Then $G - S$ contains four components. However then, $k(G - S) = 4 > 3 = |S|$ and by Theorem 6.5, G is not Hamiltonian. This sufficient condition for a graph to be non-Hamiltonian is not necessary, however; that is, there exist non-Hamiltonian graphs G such that $k(G - S) \leq |S|$ for *every* nonempty proper subset S of $V(G)$ (see Exercise 6.9).

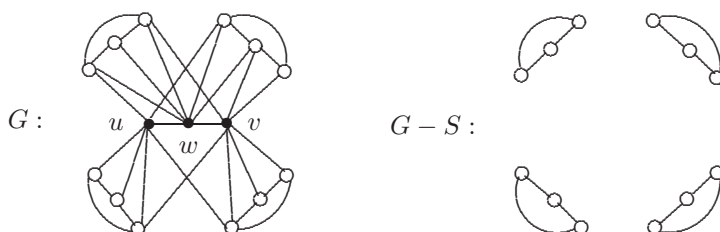


Figure 6.16: Illustrating Theorem 6.5

Although the definitions of Eulerian circuits and Hamiltonian cycles seem similar, these concepts are, in fact, quite different. While we have seen that determining whether a graph contains an Eulerian circuit is not difficult, determining whether a graph contains a Hamiltonian cycle can be exceedingly difficult. Indeed, while there is a very simple characterization of Eulerian graphs (Theorem 6.1), there is *no* such useful characterization of Hamiltonian graphs. This illustrates how making what appears to be a small change in a problem, namely from seeking an Eulerian circuit to seeking a Hamiltonian cycle, can change the problem from one that is easily solvable to one where no practical method of solution appears to exist.

As is ordinarily the case when there is no characterization of graphs possessing a certain property, one looks for sufficient conditions for a graph to have such a property. One sufficient condition for a graph to be Hamiltonian that is not difficult to apply was discovered by Oystein Ore.

Theorem 6.6 *Let G be a graph of order $n \geq 3$. If*

$$\deg u + \deg v \geq n$$

for each pair u, v of nonadjacent vertices of G , then G is Hamiltonian.

Proof. [proof by contradiction] Assume, to the contrary, that there exists a non-Hamiltonian graph G of order $n \geq 3$ such that $\deg u + \deg v \geq n$ for each pair u, v of nonadjacent vertices of G . It may be the case that if we add certain edges to G , then the resulting graph is still not Hamiltonian. Of course, if we add all possible edges to G , we obtain K_n , which is obviously Hamiltonian. Add as many edges as possible to G so that the resulting graph H is not Hamiltonian. Therefore, adding any edge to H results in a Hamiltonian graph. Also, $\deg_H u + \deg_H v \geq n$ for every pair u, v of nonadjacent vertices of H .

Since H is not complete, H contains pairs of nonadjacent vertices. Let x and y be two nonadjacent vertices of H . Thus $H + xy$ is Hamiltonian. Furthermore, every Hamiltonian cycle of $H + xy$ must contain the edge xy . This means that H contains a Hamiltonian $x - y$ path $P : x = x_1, x_2, \dots, x_n = y$. We claim that whenever $x_1 x_i$ is an edge of H , where $2 \leq i \leq n$, then $x_{i-1} x_n$ is not an edge of H (see Figure 6.17); otherwise,

$$x_1, x_i, x_{i+1}, \dots, x_n, x_{i-1}, x_{i-2}, \dots, x_1$$

is a Hamiltonian cycle of H , which is impossible. Hence for each vertex in $\{x_2, x_3, \dots, x_n\}$ that is adjacent to x_1 , there is a vertex in $\{x_1, x_2, \dots, x_{n-1}\}$ that is not adjacent to x_n . However, this means that $\deg x_n \leq (n-1) - \deg x_1$ and so

$$\deg_H x + \deg_H y \leq n - 1.$$

This is a contradiction. ■

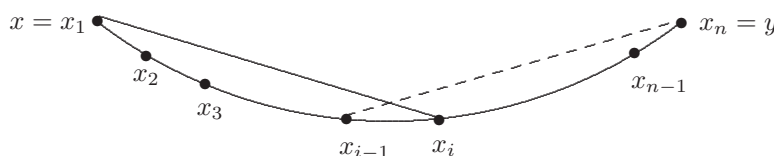
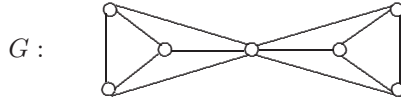


Figure 6.17: A step in the proof of Theorem 6.6

The bound given in Theorem 6.6 is sharp. For example, suppose that $n = 2k + 1 \geq 3$ is an odd integer. Let G be the graph obtained by identifying a vertex in a copy of K_{k+1} and a vertex in another copy of K_{k+1} . So $G \cong K_1 + (2K_k)$. For $n = 7$, the graph G is shown in Figure 6.18. Certainly G is non-Hamiltonian since G contains a cut-vertex. If u and v are any two nonadjacent vertices of G , then $\deg u = \deg v = k$ and so $\deg u + \deg v = 2k = n - 1$. Hence the bound given in Theorem 6.6 cannot be lowered to produce the same conclusion.

Figure 6.18: The graph $G = K_1 + (2K_3)$

There is a corollary to Theorem 6.6 due to Gabriel Dirac, whom we encountered earlier, and which is even easier to apply. Dirac was a distinguished research mathematician whose stepfather Paul Dirac was well-known for his work in quantum mechanics. In fact, Paul Dirac was awarded the Nobel Prize for physics in 1933. Gabriel Dirac was a professor of the Aarhus Universitet in Denmark.

Corollary 6.7 *Let G be a graph of order $n \geq 3$. If $\deg v \geq n/2$ for each vertex v of G , then G is Hamiltonian.*

Proof. [direct proof] Certainly, if $G \cong K_n$, then G is Hamiltonian. We may therefore assume that G is not complete. Let u and v be two nonadjacent vertices of G . Thus

$$\deg u + \deg v \geq \frac{n}{2} + \frac{n}{2} = n.$$

By Theorem 6.6, G is Hamiltonian. ■

If we look at the proof of Theorem 6.6 more carefully, then we might see that it provides a proof of another result, due to J. Adrian Bondy and Vášek Chvátal.

Theorem 6.8 *Let u and v be nonadjacent vertices in a graph G of order n such that $\deg u + \deg v \geq n$. Then $G + uv$ is Hamiltonian if and only if G is Hamiltonian.*

Proof. [direct proof, proof by contradiction] If G is a Hamiltonian graph, then certainly $G + uv$ is Hamiltonian for any two nonadjacent vertices u and v of G . Thus we need only verify the converse.

Let $G + uv$ be a Hamiltonian graph for two nonadjacent vertices u and v of a graph G , and assume, to the contrary, that G is not Hamiltonian. This implies that every Hamiltonian cycle in $G + uv$ must contain the edge uv and so G contains a Hamiltonian $u - v$ path. Since $\deg_G u + \deg_G v \geq n$, the proof of Theorem 6.6 tells us that G contains a Hamiltonian cycle. This is a contradiction. ■

The **closure** $C(G)$ of a graph G of order n is the graph obtained from G by recursively joining pairs of nonadjacent vertices whose degree sum is at least n (in the resulting graph at each stage) until no such pair remains. Figure 6.19 illustrates how to obtain the closure of a graph G , while Figure 6.20 shows three other graphs and their closures. (The graph G_3 of Figure 6.20 is the graph of Figure 6.19.) In the graph G_4 of Figure 6.20, the vertices u and v are the only two nonadjacent vertices whose degree sum is 10 (the order of G_4).

Repeated applications of Theorem 6.8 give us the following result.

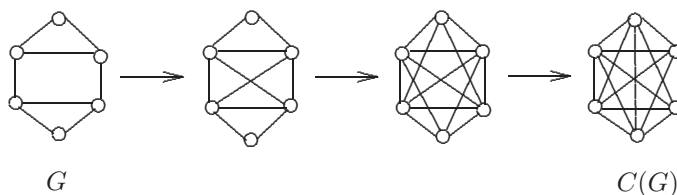


Figure 6.19: Constructing the closure of a graph

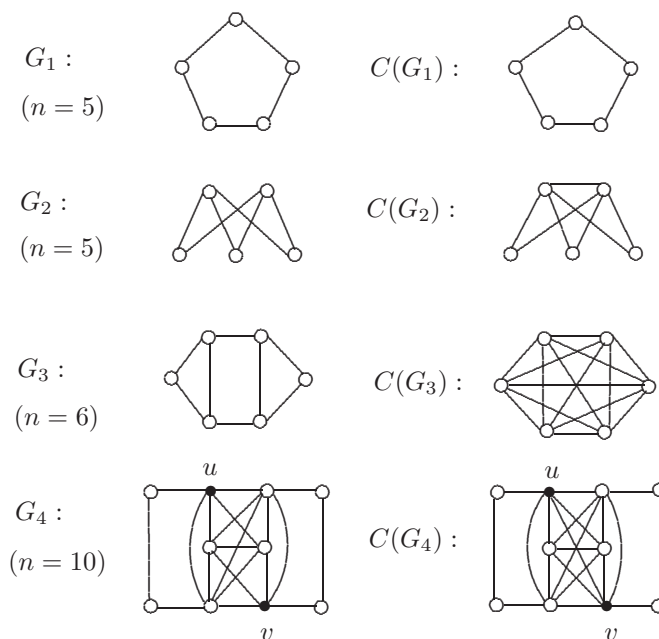


Figure 6.20: Graphs and their closures

Theorem 6.9 *A graph is Hamiltonian if and only if its closure is Hamiltonian.*

A simple but useful consequence of Theorem 6.9 is stated next.

Corollary 6.10 *If G is a graph of order at least 3 such that $C(G)$ is complete, then G is Hamiltonian.*

We saw in Corollary 6.7 (Dirac's Theorem) that if the degree of every vertex is at least $n/2$ in a graph G of order $n \geq 3$, then G is Hamiltonian. Moreover, Ore's Theorem (Theorem 6.6) requires a weaker condition to be satisfied for a graph to be Hamiltonian, as only the degree sum of every two nonadjacent vertices must be at least n . Actually, Bondy and Chvátal's Closure Theorem (Theorem 6.8) implies this as well. Indeed, Theorem 6.8 suggests that many more graphs are

Hamiltonian. Yet, it is not clear what conditions on the degrees of the vertices of a graph G of order $n \geq 3$ must be satisfied in order for Theorem 6.8 to guarantee that G is Hamiltonian. An example of such a result (which preceded Theorem 6.8 chronologically) is due to Lajos Posa. The following theorem is even more remarkable because Posa was barely a teenager when he discovered this result. Despite having a promising future in research, Posa went on to devote himself to the mathematics education of the everyday student in his native Hungary.

Theorem 6.11 *Let G be a graph of order $n \geq 3$. If for every integer j with $1 \leq j < \frac{n}{2}$, the number of vertices of G with degree at most j is less than j , then G is Hamiltonian.*

Proof. [proof by contradiction] We show that $C(G)$ is complete. Assume, to the contrary, that this is not the case. Among all pairs of nonadjacent vertices in $C(G)$, let u, w be a pair for which $\deg_{C(G)} u + \deg_{C(G)} w$ is maximum. Necessarily, $\deg_{C(G)} u + \deg_{C(G)} w \leq n - 1$. We may also assume that $\deg_{C(G)} u \leq \deg_{C(G)} w$. Let $\deg_{C(G)} u = k$. Thus $k \leq \frac{n-1}{2}$ and so

$$\deg_{C(G)} w \leq n - k - 1. \quad (6.1)$$

Let W be the set of all vertices distinct from w that are not adjacent to w . Therefore, $u \in W$. Observe that if $v \in W$, then $\deg_{C(G)} v \leq k$; for otherwise

$$\deg_{C(G)} v + \deg_{C(G)} w > \deg_{C(G)} u + \deg_{C(G)} w,$$

contradicting the defining property of the pair u, w . Therefore, the degree of every vertex of W is at most k . So by hypothesis, $|W| \leq k - 1$. Hence

$$\deg_{C(G)} w \geq (n - 1) - (k - 1) = n - k,$$

which contradicts (6.1). ■

For $j = 1$, Theorem 6.11 says that G has no vertex of degree 1. For $j = 2$, the graph G is allowed to have a vertex of degree 2. For $j = 3$, the graph G is allowed to have a vertex of degree 2 *and* a vertex of degree 3 *or* two vertices of degree 3.

Exercises for Section 6.2

6.9 We have seen that the graph G of Figure 6.10 is not Hamiltonian. Show that $k(G - S) \leq |S|$ for every nonempty proper subset S of $V(G)$. What does this say about Theorem 6.5?

6.10 Let G be a 6-regular graph of order 10 and let $u, v \in V(G)$. Prove that G , $G - v$, and $G - u - v$ are all Hamiltonian.

- 6.11 Prove that $\overline{C}_n$ is Hamiltonian for $n \geq 5$.
- 6.12 Let G be a 3-regular graph of order 12 and H a 4-regular graph of order 11.
- (a) Is $G + H$ Eulerian? Explain.
 - (b) Is $G + H$ Hamiltonian? Explain.
- 6.13 Give an example of a graph G that is
- (a) Eulerian but not Hamiltonian. (Explain why G is not Hamiltonian.)
 - (b) Hamiltonian but not Eulerian. (Explain why G is not Eulerian.)
 - (c) Hamiltonian and has an Eulerian trail but is not Eulerian.
 - (d) neither Eulerian nor Hamiltonian, but has an Eulerian trail.
- 6.14 Give an example of a graph with the following properties or explain why no such example exists:
- (a) a 2-connected (that is, connected, order at least 3, and no cut-vertices) Eulerian graph that is not Hamiltonian.
 - (b) a Hamiltonian graph G that is not Eulerian but whose complement $\overline{G}$ is Eulerian.
- 6.15 The **subdivision graph** of a graph G is that graph obtained from G by deleting *every* edge uv of G and replacing it by a vertex w of degree 2 that is joined to u and v . Is it true that if the subdivision graph of a graph G is Hamiltonian, then G is Eulerian?
- 6.16 Let G be a connected r -regular graph of even order n such that $\overline{G}$ is also connected. Show that
- (a) either G or $\overline{G}$ is Eulerian.
 - (b) either G or $\overline{G}$ is Hamiltonian.
- 6.17 For a graph G of order $n \geq 3$, the graph $G(3)$ is obtained from G by adding a new vertex v_S for each 3-element subset S of $V(G)$ and joining v_S to each vertex in S . Find all such graphs G for which $G(3)$ is Hamiltonian.
- 6.18 Show that the bound in Corollary 6.7 is sharp.
- 6.19 Let G_1 and G_2 be two graphs of order $n \geq 3$, each of which satisfies the hypothesis of Dirac's Theorem (Corollary 6.7) on Hamiltonian graphs. A graph G is constructed from $G_1 \cup G_2$ by adding edges between G_1 and G_2 such that every vertex of G_1 is joined to at least half the vertices of G_2 in such a way that every vertex of G_2 is joined to at least half the vertices of G_1 . Prove that G is Hamiltonian.

- 6.20 Let G be a graph of order $n \geq 3$ having the property that for each $v \in V(G)$, there is a Hamiltonian path with initial vertex v . Show that G is 2-connected (connected, order at least 3, and no cut-vertices) but not necessarily Hamiltonian.
- 6.21 Let G be a graph of order $n \geq 3$ such that $\deg u + \deg v \geq n - 1$ for every two nonadjacent vertices u and v . Prove that G must contain a Hamiltonian path.
- 6.22 (a) Does there exist a graph G of order 10 and size 28 that is not Hamiltonian?
- (b) Does there exist a graph H of order 10 and size 28 that is not Hamiltonian, where 8 of the 10 vertices have the degrees 5, 5, 5, 5, 5, 6, 6, 6?
- 6.23 (a) Does there exist a graph G of order $n = 2k \geq 6$ and size $m = k^2 + k - 2$ that is not Hamiltonian?
- (b) Does there exist a graph H of order $n = 2k \geq 6$ and size $m = k^2 + k - 2$ that is not Hamiltonian, where k vertices of H have degree k and $k - 2$ vertices of H have degree $k + 1$?
- 6.24 (a) A connected graph G of order $n = 2k + 1$ has $k + 1$ vertices of degree 2, no two of which are adjacent, while the remaining k vertices have degree 3 or more. Show that G is not Hamiltonian.
- (b) Give an example of a Hamiltonian graph H of order $n = 2k$ for some $k \geq 2$, where k vertices have degree 2, no two vertices of which are adjacent, while the remaining vertices have degree 3 or more.

6.3 Exploration: Hamiltonian Walks and Numbers

While certainly not every connected graph of order at least 3 contains a Hamiltonian cycle, every connected graph does contain a closed spanning walk. Indeed, if every edge of a connected graph G is replaced by two parallel edges, then the resulting multigraph M is Eulerian (see Figure 6.21). Since an Eulerian circuit in M gives rise to a closed spanning walk in G in which each edge of G appears twice, it follows that a connected graph of size m has a closed spanning walk of length $2m$ in G .

A **Hamiltonian walk** in a connected graph G is a closed spanning walk of minimum length in G . From our earlier observation, every connected graph G of size m contains a Hamiltonian walk and the length of such a walk is at most

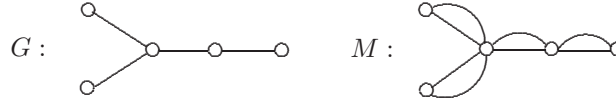


Figure 6.21: Closed spanning walks in graphs

2m. The length of a Hamiltonian walk in G is denoted by $h(G)$. Therefore, for a connected graph G of order $n \geq 3$, it follows that $h(G) = n$ if and only if G is Hamiltonian. The concept of a Hamiltonian walk was introduced by Seymour Goodman and Stephen Hedetniemi in 1973.

Although it is often difficult to determine whether a graph G is Hamiltonian, we have seen that if G satisfies any of a number of sufficient conditions, then G is Hamiltonian. However, none of these conditions is necessary and so G can be Hamiltonian without satisfying any of these conditions. In such a case, our only option may be to construct a Hamiltonian cycle in G . So the problem is reduced to finding a way to list all of the vertices of G in a cyclic sequence $v_1, v_2, \dots, v_n, v_1$ so that every pair of consecutive vertices in the sequence are adjacent. Another way to say this is to list the vertices of G in a cyclic sequence $v_1, v_2, \dots, v_n, v_1$ such that $d(v_i, v_{i+1}) = 1$ for $1 \leq i \leq n-1$ and $d(v_n, v_1) = 1$. If we also write v_1 as v_{n+1} , then the cyclic sequence $v_1, v_2, \dots, v_n, v_{n+1} = v_1$ is a Hamiltonian cycle if and only if

$$\sum_{i=1}^n d(v_i, v_{i+1}) = n.$$

Looking at Hamiltonian cycles in this manner suggests another concept. For a connected graph G of order $n \geq 3$ and a cyclic ordering

$$s : v_1, v_2, \dots, v_n, v_{n+1} = v_1$$

of $V(G)$, define the number $d(s)$ by

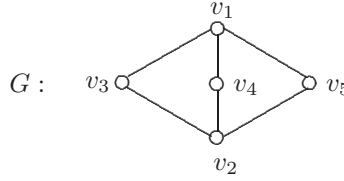
$$d(s) = \sum_{i=1}^n d(v_i, v_{i+1}).$$

Therefore, $d(s) \geq n$ for each cyclic ordering s of $V(G)$. Moreover, G is Hamiltonian if and only if there exists a cyclic ordering s' of $V(G)$ such that $d(s') = n$. The **Hamiltonian number** $h^*(G)$ of G is defined by

$$h^*(G) = \min \{d(s)\},$$

where the minimum is taken over all cyclic orderings s of $V(G)$. For the graph $G \cong K_{2,3}$ of Figure 6.22, $d(s_1) = 8$ and $d(s_2) = 6$ for the cyclic orderings

$$s_1 : v_1, v_2, v_3, v_4, v_5, v_1 \text{ and } s_2 : v_1, v_3, v_2, v_4, v_5, v_1$$

Figure 6.22: A graph G with $h^*(G) = 6$

of $V(G)$. Since G is a non-Hamiltonian graph of order 5 and $d(s_2) = 6$, it follows that $h^*(G) = 6$.

We are about to see that there is an alternative way to define the length $h(G)$ of a Hamiltonian walk in G . Denote the length of a walk W by $L(W)$.

Theorem 6.12 *For every connected graph G ,*

$$h^*(G) = h(G).$$

Proof. First, we show that $h(G) \leq h^*(G)$. Let $s : v_1, v_2, \dots, v_n, v_{n+1} = v_1$ be a cyclic ordering of $V(G)$ for which $d(s) = h^*(G)$. For each integer i with $1 \leq i \leq n$, let P_i be a $v_i - v_{i+1}$ geodesic in G . Thus $L(P_i) = d(v_i, v_{i+1})$. The union of the paths P_i form a Hamiltonian walk W in G . Therefore,

$$h(G) \leq L(W) = \sum_{i=1}^n L(P_i) = \sum_{i=1}^n d(v_i, v_{i+1}) = d(s) = h^*(G).$$

Next, we show that $h^*(G) \leq h(G)$. Let W be a Hamiltonian walk in G with $L(W) = h(G)$. Suppose that $W : x_1, x_2, \dots, x_N, x_1$, where then $N \geq n$. Define $v_1 = x_1$ and $v_2 = x_2$. For $3 \leq i \leq n$, define v_i to be x_{j_i} , where j_i is the smallest positive integer such that $x_{j_i} \notin \{v_1, v_2, \dots, v_{i-1}\}$. Then $s : v_1, v_2, \dots, v_n, v_{n+1} = v_1$ is a cyclic ordering of $V(G)$. For each i with $1 \leq i \leq n$, let Q_i be the $v_i - v_{i+1}$ subwalk of W and so $d(v_i, v_{i+1}) \leq L(Q_i)$. Since

$$h^*(G) \leq \sum_{i=1}^n d(v_i, v_{i+1}) \leq \sum_{i=1}^n L(Q_i) \leq L(W) = h(G),$$

we have the desired result. ■

As a consequence of Theorem 6.12, we can denote the Hamiltonian number of a graph G by $h(G)$, which is then the length of Hamiltonian walk in G .

For the graph $G \cong K_{2,3}$ of Figure 6.22 and the cyclic orderings $s_1 : v_1, v_2, v_3, v_4, v_5, v_1$ and $s_2 : v_1, v_3, v_2, v_4, v_5, v_1$ of $V(G)$, we saw that $d(s_1) = 8$ and $d(s_2) = 6$. Actually, it is not difficult to show that $d(s)$ is either 8 or 6 for *every* cyclic ordering s of $V(G)$. This suggests another parameter of a connected graph. The **upper Hamiltonian number** $h^+(G)$ of a connected graph G is defined as

$$h^+(G) = \max \{d(s)\},$$

where the maximum is taken over all cyclic orderings s of $V(G)$. Therefore, $h^+(G) = 8$ for the graph G of Figure 6.22.

As another example, we consider the Petersen graph. Label the vertices of the Petersen graph PG as shown in Figure 6.23. Since PG is a non-Hamiltonian graph of order 10, $h(PG) \geq 11$. On the other hand, let $s : x_1, x_2, \dots, x_{11} = x_1$ be any cyclic ordering of PG . Since $\text{diam}(PG) = 2$, it follows that $d(x_i, x_{i+1}) \leq 2$ for $1 \leq i \leq 10$. Hence $d(s) \leq 2 \cdot 10 = 20$ and so $h^+(PG) \leq 20$. Therefore,

$$11 \leq h(PG) < h^+(PG) \leq 20.$$

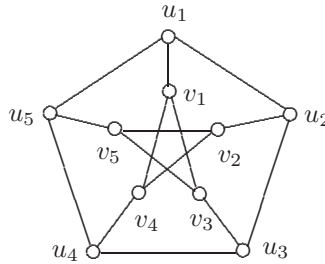


Figure 6.23: The Petersen graph

In fact, $h(PG) = 11$ and $h^+(PG) = 20$. Moreover, consider the sequences s_i ($1 \leq i \leq 10$):

$$\begin{aligned} s_1 &: u_1, u_2, u_3, u_4, u_5, v_5, v_2, v_4, v_3, v_1, u_1 \\ s_2 &: u_1, u_2, u_3, u_4, u_5, v_5, v_2, v_3, v_4, v_1, u_1 \\ s_3 &: u_1, u_2, u_3, u_5, u_4, v_4, v_2, v_3, v_5, v_1, u_1 \\ s_4 &: u_1, u_3, u_5, u_2, u_4, v_4, v_2, v_5, v_3, v_1, u_1 \\ s_5 &: u_1, u_3, u_5, u_2, u_4, v_3, v_5, v_2, v_4, v_1, u_1 \\ s_6 &: u_1, u_3, u_5, u_2, u_4, v_5, v_2, v_4, v_3, v_1, u_1 \\ s_7 &: u_1, u_3, u_5, u_2, u_4, v_3, v_5, v_4, v_2, v_1, u_1 \\ s_8 &: u_1, u_3, u_5, u_2, v_2, u_4, v_3, v_4, v_5, v_1, u_1 \\ s_9 &: u_1, u_3, u_5, u_2, u_4, v_2, v_3, v_4, v_5, v_1, u_1 \\ s_{10} &: u_1, u_3, u_5, u_2, u_4, v_1, v_2, v_3, v_4, v_5, u_1 \end{aligned}$$

Since $d(s_i) = 10 + i$ for $1 \leq i \leq 10$, it follows that for each integer k with $11 \leq k \leq 20$, there exists a cyclic ordering s of $V(PG)$ such that $d(s) = k$.

Exercises for Section 6.3

- 6.25 For $G_1 \cong K_n$ and $G_2 \cong K_{s,t}$, where $n \geq 3$ and $1 \leq s \leq t$, find $h(G_i)$ and $h^+(G_i)$ for $i = 1, 2$.

- 6.26 Give an example of a graph G of order $n \geq 3$ such that $h(G) = n + 1$. Verify that your example is correct.
- 6.27 Give an example of a graph G of order $n \geq 3$ such that $h(G) = 2n - 2$. Verify that your example is correct.
- 6.28 Let C_n be a cycle of order $n \geq 3$.
- (a) What is $h^+(C_n)$ if n is even?
 - (b) What can you say about $h^+(C_n)$ if n is odd?
- 6.29 (a) Prove that if G is a connected graph of order n and diameter $d \geq 2$, then $h^+(G) \leq nd$.
- (b) Is the upper bound in (a) is sharp?
- 6.30 Determine all graphs G for which $h(G) = h^+(G)$.
- 6.31 Instead of considering cyclic sequences of the vertices of a graph G (and studying the Hamiltonian and upper Hamiltonian numbers), consider linear sequences of the vertices of G .
- 6.33 Ask and answer a question of your own concerning Hamiltonian numbers and/or upper Hamiltonian numbers.

6.4 Excursion: The Early Books of Graph Theory

Theorem 6.1, published in 1736, is considered to be the beginning of graph theory. It wasn't until 1936, however, when the first textbook (in German) was written on graph theory by Dénes König. König was born in Budapest, Hungary on September 21, 1884. König was interested in mathematics at an early age, no doubt influenced by his father who was a well-known mathematics professor. Indeed, he even published a paper as a high school student.

König spent nine semesters doing university work, the first four in Budapest and the last five in Gottingen. He attended lectures by Hermann Minkowski on analysis situs, which is what topology was called in its early days. The fact that Minkowski was interested in graph theory played a big role in the mathematics König decided to work on. König received his doctorate in 1907 and wrote his dissertation in geometry. He acquired a faculty position at the Technische Hochschule in Budapest that year and remained a member of the faculty there until his death. Among the courses König taught was graph theory, although the name "graph theory" never appeared in the catalogue at the university until 1927. It fell under the heading of analysis situs prior to that time. In 1935 König

became a full professor. König was well-known for his enthusiastic lectures, although his lectures were not always well attended. Because of him, however, a number of dedicated students were introduced to this new area of mathematics. Indeed, under his influence, Hungarian researchers turned to this field, including Paul Erdős, Tibor Gallai, and Paul Turán.

Although an excellent mathematician, König's main accomplishment is probably the popularization and recognition of graph theory. Because of his efforts, graph theory grew from being a collection of isolated results in a branch of recreational mathematics to a recognized new area of the mathematical sciences. Although he was belittled by some mathematicians, he was not discouraged. He believed in the future of graph theory. Indeed, König would often begin a lecture on graph theory by saying:

Graph theory is one of the most interesting of mathematical disciplines.

In 1936, König's book *Theorie der endlichen und unendlichen Graphen* (the very first book ever written solely on graph theory) was published in Leipzig although Oswald Veblen had discussed graph theory in his book *Analysis Situs*, the first book written on topology. König worked on his book with great care for many years. His book awakened the interest of many young mathematicians in graph theory, although its impact was only felt after World War II. In 1944, after the occupation of Hungary by the Nazis, König worked to assist persecuted mathematicians. Rather than be persecuted himself, he committed suicide on October 19, 1944. In 1950 König's book was reprinted in the United States. For over 20 years, König's work was the only book on graph theory, until 1958 when *Théorie des Graphes et Ses Applications* by Claude Berge was published.

Claude Berge was born on June 5, 1926. He was the one individual to spread the word of graph theory throughout France. Despite König's book, prior to the 1950s many mathematicians thought little of combinatorics and graph theory. Most French mathematicians tended to resist graph theory and preferred the term "network" instead. However, because of Berge's efforts, all this changed.

When graph theory was introduced to Berge, it was a subject that was unknown in France. It was Berge's intent to make some sense of this new field. At first, he worked on graph theory just for himself, but in 1958 he published his book on graph theory (others would come later). He found the subject interesting and alive with many applications. However, it wasn't only the French to whom Berge introduced graph theory. He traveled widely and lectured on the subject. Many of Berge's works were translated into other languages. In 1993 Berge was awarded the Euler Medal by the Institute of Combinatorics and Its Applications.

Berge made contributions to other areas of mathematics, including game theory and topology. He introduced an alternative to the Nash equilibrium (named after John Nash whose life was chronicled in the 2002 academy award winning movie *A Beautiful Mind*) called the Berge equilibrium. Berge also coined the term *hypergraph*.

Berge had many interests besides mathematics. He had a special affinity for Chinese works of art and was a skillful chess player. Berge loved to write. In 1994 he authored a mathematical murder mystery, titled *Who Killed the Duke of Densmore?* in which the detective investigates the murder of the Duke of Densmore and uses graphs (actually interval graphs, which we do not discuss here) to find the murderer. Berge died on June 30, 2002.

The next major book on graph theory, titled *Theory of Graphs*, was published in 1962 by the American Mathematical Society, four years after Berge's book. Its author, Oystein Ore (pronounced Austin OR-ah) was born in Oslo, Norway in 1899. Ore attended the University of Oslo, from which he graduated in 1922. He received his Ph.D. in 1924. After this, he spent time in Paris and Gottingen and then returned to the University of Oslo. In 1927 he went to the United States where he became a professor of mathematics at Yale University. Ore wrote over a hundred papers and a number of books.

Ore was well-known for his work in algebra and number theory before he started working in graph theory. He had much to do with introducing graph theory to the English-speaking world with his 1962 book. Ore once wrote, in reference to Euler's solution to the Königsberg Bridge Problem, that

The theory of graphs is one of the few fields of mathematics with a definite birth date.

Ore died during the summer of 1968, only a few months before Ore was to attend and be the principal speaker at the first of nine quadrennial graph theory conferences to be held at Western Michigan University.

Although some books dealing with special topics in graph theory and applications of graph theory were published during 1959–1969, only two other major books on graph theory were published before 1970, both in 1969. One of these was the 1969 book *Teoriia Konechnykh Grafov (Theory of Finite Graphs)* by Alexander A. Zykov, who has had the greatest influence on the development of and interest in graph theory in Russia. Born in Kiev in 1922, Zykov was introduced to graph theory during 1943–44 while he was a student at Gor'kiy State University. He was in Novosibirsk during 1959–69 and organized a graph theory seminar at the Mathematical Institute of the Siberian branch of the Academy Sciences in the USSR. Since 1969 Zykov has been at Odessa State Polytechnic University in Ukraine, where he has been organizer of the Odessa seminar in discrete mathematics.

The second book titled *Graph Theory* was written by Frank Harary. Harary was born on March 11, 1921 in New York City. He received his Ph.D. in 1948 from the University of California in Berkeley. Harary went to the University of Michigan, where he was a faculty member until 1986. Since then he has been a faculty member at New Mexico State University.

While working with social psychologists at the University of Michigan, he independently discovered graph theory. Harary has spent much of his life traveling and lecturing on graph theory, thereby introducing this subject to many mathematicians around the world. Along the way, he acquired numerous coauthors,

which led to hundreds of publications on all aspects of the subject. Harary, known for his lucid writing style and his enthusiasm for the subject, has been referred to as the *Ambassador of Graph Theory*.

On the personal side, Harary, an avid devotee of both folk dancing and classical music, is also a “man of lists”, often compiling lists of such things as the cities and universities where he has lectured, journals in which he has published, and coauthors he has had.

At one time Alexander Zykov gave Frank Harary a framed set of three pictures of lions, namely (1) a lion sleeping, (2) a lion awakening, and (3) a lion roaring to represent (1) graph theory yesterday, (2) graph theory today, and (3) graph theory tomorrow. Although meant as a comical gift, Zykov indicated that if one were to define “yesterday” as “before 1936”, “today” as “1936–1970”, and “tomorrow” as “after 1970”, then there is a certain amount of truth to this representation of graph theory.

William T. Tutte (1917–2002), a mathematician who made numerous significant contributions to graph theory, was a shy man with a clever sense of humor. Tutte loved to write poetry, often under the pseudonym of Blanche Descartes. As Descartes, he reflected on graph theory in his 1969 poem titled “The Expanding Unicurse”:

*Some citizens of Königsberg
Were walking on the strand
Beside the river Pregel
With its seven bridges spanned.
“O Euler, come and walk with us,”
Those burghers did beseech.
“We’ll roam the seven bridges o’er,
And pass but once by each.”
“It can’t be done,” thus Euler cried.
“Here comes the Q. E. D.
Your islands are but vertices,
And four have odd degree.”
From Königsberg to König’s book,
So runs the graphic tale,
And still it grows more colorful,
In Michigan and Yale.*

The mention of Michigan and Yale in the last line of the poem refers to the universities of Frank Harary and Oystein Ore, respectively. We will encounter William Tutte again.